

# 1

## Matrices

### 1.0 INTRODUCTION

The concept of matrices and their basic operations were introduced by the British mathematician Arthur Cayley in the year 1858. He wondered whether this part of mathematics will ever be used. However, after 67 years, in 1925, the German physicist Heisenberg used the algebra of matrices in his revolutionary theory of quantum mechanics. Over the years, matrices have been found as an elegant and powerful tool in almost all branches of Science and Engineering like electrical networks, graph theory, optimisation techniques, system of differential equations, stochastic processes, computer graphics, etc. Because of the digital computers, usage of matrix methods have become greatly fruitful.

Here we study certain numbers associated with a square matrix, called eigen values and certain vectors associated with them, called eigen vectors. These are useful in the study of canonical forms of a matrix such as diagonalisation and in the study of quadratic forms. The problem of determining eigen values and eigen vectors of a square matrix is called an **eigen value problem**.

We shall now introduce some basic concepts.

### 1.1 VECTOR

The vector  $2\vec{i} + 3\vec{j} - 4\vec{k}$  can be regarded as the triplet  $(2, 3, -4)$ .

**Definition 1.1.1** An ordered  $n$ -tuple  $(x_1, x_2, \dots, x_n)$  of numbers  $x_1, x_2, \dots, x_n$  is called an  $n$ -dimensional vector.

For example the triplet  $(2, 3, -4)$  is a 3-dimensional vector.  $(1, 0, -2, 3)$  is a 4-dimensional vector.

A row matrix is also called a row vector and a column matrix is called a column vector.

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**Definition 1.1.2** If  $X_1 = (a_1, a_2, \dots, a_n)$ ,  $X_2 = (b_1, b_2, \dots, b_n)$  be two  $n$ -dimensional vectors, then their sum and scalar multiplications are

$$\begin{aligned} X_1 + X_2 &= (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n), \\ \alpha X_1 &= (\alpha a_1, \alpha a_2, \dots, \alpha a_n), \text{ which are } n\text{-dimensional vectors.} \\ X_1 &= X_2 \text{ if and only if } a_1 = b_1, a_2 = b_2, \dots, a_n = b_n \end{aligned}$$

**Definition 1.1.3 Linear combination** If  $X_1, X_2, \dots, X_r$  are  $r$  vectors of  $n$ -dimension and if  $\alpha_1, \alpha_2, \dots, \alpha_r$  are numbers, then  $\alpha_1 X_1 + \alpha_2 X_2 + \dots + \alpha_r X_r$  is called a linear combination of the vectors  $X_1, X_2, \dots, X_r$ .

### Definition 1.1.4 Linearly dependent and independent vectors

- (a) The set of vectors  $X_1, X_2, \dots, X_r$  is said to be linearly dependent if there exist numbers  $\alpha_1, \alpha_2, \dots, \alpha_r$ , not all zero, such that  $\alpha_1 X_1 + \alpha_2 X_2 + \dots + \alpha_r X_r = 0$
- (b) The set of vectors  $X_1, X_2, \dots, X_r$  is said to be linearly independent if any relation of the form  $\alpha_1 X_1 + \alpha_2 X_2 + \dots + \alpha_r X_r = 0 \Rightarrow \alpha_1 = 0, \alpha_2 = 0, \dots, \alpha_r = 0$

### Note

- (i) If  $X_1, X_2, \dots, X_r$  are linearly dependent, then some vector is a linear combination of others.
- (ii) In a plane or 2-dimensional space, non-collinear vectors are linearly independent vectors whereas collinear vectors are dependent vectors.  
In 3-dimensional space, non-coplanar vectors are linearly independent vectors whereas coplanar vectors are dependent vectors.  
e.g.,  $\vec{i} = (1, 0, 0), \vec{j} = (0, 1, 0), \vec{k} = (0, 0, 1)$  are linearly independent vectors.
- (iii) Any set of vectors containing zero vector  $0$  is a linearly dependent set.
- (iv) Rank of an  $m \times n$  matrix  $A$  is equal to the maximum number of independent column vectors or row vectors of  $A$ .
- (v) **A useful result to test linear independence:**

Let  $X_1, X_2, \dots, X_n$  be  $n$  vectors of  $n$ -dimensional space. Let  $A$  be the matrix having these  $n$ -vectors as columns (or rows). Then  $A$  is a square matrix of order  $n$ . If  $|A| \neq 0$ , then  $X_1, X_2, \dots, X_n$  are linearly independent.  
If  $|A| = 0$ , then  $X_1, X_2, \dots, X_n$  are linearly dependent.

## WORKED EXAMPLES

### Example 1

Show that the vectors  $(1, 2, 3), (3, -2, 1), (1, -6, -5)$  are linearly dependent.  
[MU 1996]

**Solution**

Let  $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & -2 & -6 \\ 3 & 1 & -5 \end{bmatrix}$  with the vectors as columns.

$$\begin{aligned} \text{then } |A| &= \begin{vmatrix} 1 & 3 & 1 \\ 2 & -2 & -6 \\ 3 & 1 & -5 \end{vmatrix} \\ &= 1 \cdot (10 + 6) - 3(-10 + 18) + 1 \cdot (2 + 6) \\ &= 16 - 24 + 8 = 0 \end{aligned}$$

$\therefore$  the vectors  $(1, 2, 3)$ ,  $(3, -2, 1)$  and  $(1, -6, -5)$  are linearly dependent.

**Example 2**

Show that the vectors  $X_1 = (1, 2, -3, 4)$ ,  $X_2 = (3, -1, 2, 1)$ ,  $X_3 = (1, -5, 8, -7)$  are linearly dependent and find the relation between them.

**Solution**

Let  $A = \begin{bmatrix} 1 & 2 & -3 & 4 \\ 3 & -1 & 2 & 1 \\ 1 & -5 & 8 & -7 \end{bmatrix}$  with the vectors as rows.

We shall use elementary row operations.

$$\begin{aligned} \therefore A &\sim \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 11 & -11 \\ 0 & -7 & 11 & -11 \end{bmatrix} & \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 = R'_2 \\ R_3 \rightarrow R_3 - R_1 = R'_3 \end{array} \\ &\sim \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 11 & -11 \\ 0 & 0 & 0 & 0 \end{bmatrix} & R_3 \rightarrow R'_3 - R'_2 = R''_3 \end{aligned}$$

Since the maximum number of non-zero rows is 2, which is less than the number of vectors, the given vectors are linearly dependent.

The relation between them is obtained as below.

$$\begin{aligned} R''_3 = 0 &\Rightarrow R'_3 - R'_2 = 0 \\ &\Rightarrow R_3 - R_1 - (R_2 - 3R_1) = 0 \\ &\Rightarrow R_3 - R_2 + 2R_1 = 0 \end{aligned}$$

Since the rows are vectors, we get  $X_3 - X_2 + 2X_1 = 0$  which is the relation between the vectors.

**Note** The rows of the matrix are the given vectors. So, only row operations must be used to find the relationship between the vectors.

## 1.2 EIGEN VALUES AND EIGEN VECTORS

**Definition 1.2.1** Let  $A$  be a square matrix of order  $n$ . A number  $\lambda$  is called an **eigen value** of  $A$  if there exists a non-zero column matrix  $X$  such that  $AX = \lambda X$ . Then  $X$  is called an eigen vector of  $A$  corresponding to  $\lambda$ .

$$\text{If } A = [a_{ij}]_{n \times n} \text{ and } X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \text{ then}$$

$AX = \lambda X \Rightarrow (A - \lambda I)X = 0$ . This will represent a system of linear homogeneous equations in  $x_1, x_2, \dots, x_n$ . Since  $X \neq 0$  at least one of the  $x_i \neq 0$ . Hence the homogeneous system has nontrivial solutions.

$\therefore$  the determinant of coefficients  $|A - \lambda I| = 0$ . This equation is called the **characteristic equation** of  $A$ . The determinant  $|A - \lambda I|$ , on expansion, will be a  $n^{\text{th}}$  degree polynomial in  $\lambda$  and is known as the **characteristic polynomial of  $A$** . The roots of the characteristic equation are the **eigen values of  $A$** .

**Definition 1.2.2 Characteristic equation and characteristic polynomial** If  $\lambda$  is a characteristic root of a square matrix  $A$ , then  $|A - \lambda I| = 0$  is called the characteristic equation of  $A$ .

The polynomial  $|A - \lambda I|$  in  $\lambda$  is called the characteristic polynomial of  $A$ .

### Note

- (1) The word 'eigen' is German, which means 'characteristic' or 'proper'. So, an eigen value is also known as characteristic root or proper value. Sometimes it is also known as latent root.

- (2) If  $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ ,  $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ , then the characteristic equation of  $A$  is

$$\begin{aligned} & |A - \lambda I| = 0 \\ \Rightarrow & \begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} = 0 \\ \Rightarrow & (a_{11} - \lambda)(a_{22} - \lambda) - a_{21}a_{12} = 0 \\ \Rightarrow & a_{11} \cdot a_{22} - (a_{11} + a_{22})\lambda + \lambda^2 - a_{21}a_{12} = 0 \\ \Rightarrow & \lambda^2 - (a_{11} + a_{22})\lambda + (a_{11}a_{22} - a_{21}a_{12}) = 0 \\ \Rightarrow & \lambda^2 - S_1\lambda + S_2 = 0 \end{aligned}$$

where  $S_1 = a_{11} + a_{22} = \text{sum of the diagonal elements of } A$ .

$$S_2 = a_{11} \cdot a_{22} - a_{21} \cdot a_{12} = |A|$$

(3) If  $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ ,  $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ , then the characteristic equation of A is

$$\Rightarrow \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix} = 0$$

Expanding this determinant we will get

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0,$$

where  $S_1$  = sum of the diagonal elements of A  
 $S_2$  = sum of the minors of elements of the main diagonal  
 $S_3 = |A|$

We will use this formula in problems.

**Definition 1.2.3** The set of all distinct eigen values of the square matrix  $A$  is called the **spectrum of A**. The largest of the absolute values of the eigen values of  $A$  is called the **spectral radius of A**. The set of all eigen vectors corresponding to an eigen value  $\lambda$  of  $A$ , together with zero vector, forms a vector space which is called the **eigenspace** of  $A$  corresponding to  $\lambda$ .

## Properties of eigen vectors

### Theorem 1.2.1

- (1) Eigen vector corresponding to an eigen value is not unique.
- (2) Eigen vectors corresponding to different eigen values are linearly independent.

**Proof** (1) Let  $\lambda$  be an eigen value of a square matrix  $A$  of order  $n$ .

Let  $X$  be an eigen vector corresponding to  $\lambda$ .

Then  $AX = \lambda X$

Multiply by a constant  $C$

$$\therefore C(AX) = C(\lambda X)$$

$$\Rightarrow A(CX) = \lambda(CX)$$

Since  $C \neq 0, X \neq 0$  we have  $CX \neq 0$

$\therefore CX$  is an eigen vector corresponding to  $\lambda$  for any  $C \neq 0$ . Hence eigen vector is not unique for the eigen value  $\lambda$ .

(2) Let  $\lambda_1, \lambda_2$  be two different eigen values of  $A$ .

Let  $X_1, X_2$  be corresponding eigen vectors.

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$$\therefore AX_1 = \lambda_1 X_1 \quad (1)$$

$$\text{and} \quad AX_2 = \lambda_2 X_2 \quad (2)$$

We have to prove  $X_1$  and  $X_2$  are linearly independent.

$$\text{Suppose} \quad \alpha_1 X_1 + \alpha_2 X_2 = 0 \quad (3)$$

$$\text{then} \quad A(\alpha_1 X_1 + \alpha_2 X_2) = 0$$

$$\Rightarrow \quad \alpha_1 (AX_1) + \alpha_2 (AX_2) = 0$$

$$\Rightarrow \quad \alpha_1 (\lambda_1 X_1) + \alpha_2 (\lambda_2 X_2) = 0$$

$$\Rightarrow \quad (\alpha_1 \lambda_1) X_1 + (\alpha_2 \lambda_2) X_2 = 0 \quad (4)$$

Multiply (3) by  $\lambda_1$ , we get

$$\lambda_1 (\alpha_1 X_1) + \lambda_1 (\alpha_2 X_2) = 0$$

$$\Rightarrow \quad (\alpha_1 \lambda_1) X_1 + (\alpha_2 \lambda_1) X_2 = 0 \quad (5)$$

$$(4) - (5) \Rightarrow \quad \alpha_2 (\lambda_2 - \lambda_1) X_2 = 0 \quad (6)$$

Since  $\lambda_1 \neq \lambda_2 \Rightarrow \lambda_2 - \lambda_1 \neq 0$  and  $X_2 \neq 0$

$$\therefore \quad (\lambda_2 - \lambda_1) X_2 \neq 0$$

$$\therefore (6) \Rightarrow \alpha_2 = 0 \quad \therefore (3) \Rightarrow \alpha_1 X_1 = 0$$

$$\Rightarrow \quad \alpha_1 = 0 \quad \text{since } X_1 \neq 0.$$

$$\text{Thus } \alpha_1 X_1 + \alpha_2 X_2 = 0 \quad \Rightarrow \quad \alpha_1 = 0 \text{ and } \alpha_2 = 0$$

$\therefore X_1$  and  $X_2$  are linearly independent. ■

### Note

- (1) If all the  $n$  eigen values  $\lambda_1, \lambda_2, \dots, \lambda_n$  of  $A$  are different, then the corresponding eigen vectors  $X_1, X_2, \dots, X_n$  are linearly independent.
- (2) A given eigen vector of  $A$  corresponds to only one eigen value of  $A$ .
- (3) Eigen vectors corresponding to equal eigen values may be linearly independent or dependent.

## WORKED EXAMPLES

### Example 3

Find the eigen values and eigen vectors of the matrix  $\begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$ .

[MU 1997]

**Solution**

Let  $A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$ . The characteristic equation of A is

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 4-\lambda & 1 \\ 3 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^2 - S_1\lambda + S_2 = 0$$

where  $S_1$  = sum of the diagonal elements of  $A = 4 + 2 = 6$

$$S_2 = |A| = \begin{vmatrix} 4 & 1 \\ 3 & 2 \end{vmatrix} = 8 - 3 = 5$$

$\therefore$  the characteristic equation is  $\lambda^2 - 6\lambda + 5 = 0$

$$\Rightarrow (\lambda - 1)(\lambda - 5) = 0 \Rightarrow \lambda = 1, 5$$

which are the eigen values of A.

**To find eigen vectors:**

Let  $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$  be an eigen vector of A corresponding to  $\lambda$ .

$$\text{Then } (A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} 4-\lambda & 1 \\ 3 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} (4-\lambda)x_1 + x_2 = 0 \\ 3x_1 + (2-\lambda)x_2 = 0 \end{cases} \quad (I)$$

**Case (i)** If  $\lambda = 1$ , then equations (I) become

$$3x_1 + x_2 = 0$$

$$3x_1 + x_2 = 0$$

$$\therefore x_2 = -3x_1$$

Choosing  $x_1 = 1$ , we get  $x_2 = -3$

$$\therefore \text{eigen vector is } X_1 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

**Case (ii)** If  $\lambda = 5$ , then equations (I) become

$$-x_1 + x_2 = 0 \text{ and } 3x_1 - 3x_2 = 0$$

$$\therefore x_1 = x_2$$

Choosing  $x_1 = 1$ , we get  $x_2 = 1$

$$\therefore \text{eigen vector is } X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

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Thus eigen values of A are 1, 5 and the corresponding eigen vectors are  $\begin{bmatrix} 1 \\ -3 \end{bmatrix}$ ,  $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

**Note:** In case (i) we have only one equation  $3x_1 + x_2 = 0$  to solve for  $x_1$  and  $x_2$ . So, we have infinite number of solutions  $x_1 = k$ ,  $x_2 = -3k$ , for any  $k \neq 0$ .

We have chosen the simplest solution. In fact  $\begin{bmatrix} k \\ -3k \end{bmatrix} = k \begin{bmatrix} 1 \\ -3 \end{bmatrix}$  is an eigen vector for  $\lambda = 1$  for any  $k \neq 0$ . So, for  $\lambda = 1$  there are many eigen vectors. This verifies property 1.

### Example 4

Show that the real matrix  $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$  has two eigen vectors  $\begin{bmatrix} 1 \\ i \end{bmatrix}$  and  $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ , where  $b \neq 0$ .

### Solution

$$\text{Let } A = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}.$$

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} a - \lambda & b \\ -b & a - \lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^2 - S_1\lambda + S_2 = 0$$

$$\text{where } S_1 = a + a = 2a$$

$$S_2 = |A| = \begin{vmatrix} a & b \\ -b & a \end{vmatrix} = a^2 + b^2$$

$$\therefore \text{ the characteristic equation is } \lambda^2 - 2a\lambda + (a^2 + b^2) = 0$$

$$\begin{aligned} \Rightarrow \lambda &= 2a \pm \sqrt{4a^2 - 4(a^2 + b^2)} \\ &= \frac{2a \pm \sqrt{-4b^2}}{2} \\ &= \frac{2a \pm 2ib}{2} \\ &= a + ib \text{ or } a - ib \end{aligned}$$

which are the eigen values of A.

### To find eigen vectors

Let  $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$  be an eigen vector of A corresponding to  $\lambda$ .



$$\begin{aligned} \text{Then } (A - \lambda I)X = 0 &\Rightarrow \begin{bmatrix} a - \lambda & b \\ -b & a - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \Rightarrow &\left. \begin{aligned} (a - \lambda)x_1 + bx_2 &= 0 \\ -bx_1 + (a - \lambda)x_2 &= 0 \end{aligned} \right\} \quad (I) \end{aligned}$$

**Case (i)** If  $\lambda = a + ib$ , then the equations (I) become

$$(a - (a + ib))x_1 + bx_2 = 0$$

and

$$-bx_1 + (a - a - ib)x_2 = 0$$

$\Rightarrow$

$$-ibx_1 + bx_2 = 0 \quad \Rightarrow \quad x_2 = ix_1$$

and

$$-bx_1 - ibx_2 = 0 \quad \Rightarrow \quad -x_1 = ix_2$$

$$\Rightarrow \quad i^2 x_1 = ix_2 \quad \Rightarrow \quad x_2 = ix_1$$

So, we have only one equation  $x_2 = ix_1$

Choosing  $x_1 = 1$ , we get  $x_2 = i$

$\therefore$  an eigen vector is  $X_1 = \begin{bmatrix} 1 \\ i \end{bmatrix}$

**Case (ii)** If  $\lambda = a - ib$ , then the equations (I) become

$$(a - a + ib)x_1 + bx_2 = 0$$

and

$$-bx_1 + (a - a + ib)x_2 = 0$$

$\Rightarrow$

$$ibx_1 + bx_2 = 0 \quad \Rightarrow \quad x_2 = -ix_1$$

$$-bx_1 + ibx_2 = 0 \quad \Rightarrow \quad x_2 = -ix_1$$

Choosing  $x_1 = 1$ , we get  $x_2 = -i$

$\therefore$  an eigen vector is  $X_2 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$

Thus the eigen values of  $A$  are  $a + ib$ ,  $a - ib$  and the corresponding eigen vectors are  $X_1 = \begin{bmatrix} 1 \\ i \end{bmatrix}$ ,  $X_2 = \begin{bmatrix} 1 \\ -i \end{bmatrix}$

### Example 5

Find the eigen values and eigen vectors of the matrix  $\begin{bmatrix} 3 & -4 & 4 \\ 1 & -2 & 4 \\ 1 & -1 & 3 \end{bmatrix}$ .

**Solution**

$$\text{Let } A = \begin{bmatrix} 3 & -4 & 4 \\ 1 & -2 & 4 \\ 1 & -1 & 3 \end{bmatrix}$$

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The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 3-\lambda & -4 & 4 \\ 1 & -2-\lambda & 4 \\ 1 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1$  = sum of main diagonal elements of A  
 $= 3 + (-2) + 3 = 4$

$S_2$  = sum of minors of diagonal elements of A

$$\begin{aligned} &= \begin{vmatrix} -2 & 4 \\ -1 & 3 \end{vmatrix} + \begin{vmatrix} 3 & 4 \\ 1 & 3 \end{vmatrix} + \begin{vmatrix} 3 & -4 \\ 1 & -2 \end{vmatrix} \\ &= -6 + 4 + 9 - 4 + (-6) + 4 = -2 + 5 + (-2) = 1 \end{aligned}$$

$$S_3 = |A| = 3(-6 + 4) + 4(3 - 4) + 4(-1 + 2) = -6 - 4 + 4 = -6$$

$\therefore$  The characteristic equation is  $\lambda^3 - 4\lambda^2 + \lambda + 6 = 0$

We choose integer factors of constant term 6 for trial solution. We find  $\lambda = -1$  is a root. To find the other roots we perform synthetic division

Other roots are given by

$$\begin{aligned} &\lambda^2 - 5\lambda + 6 = 0 & -1 \bigg| & \begin{array}{cccc} 1 & -4 & 1 & 6 \\ 0 & -1 & 5 & -6 \\ \hline & 1 & -5 & 6 \end{array} \\ \Rightarrow &(\lambda - 2)(\lambda - 3) = 0 \\ \Rightarrow &\lambda = 2 \text{ or } 3 \end{aligned}$$

$\therefore$  the eigen values are  $\lambda = -1, 2, 3$  [different roots]

**To find eigen vectors:**

Let  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector corresponding to the eigen value  $\lambda$ .

$$\text{Then } (A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} 3-\lambda & -4 & 4 \\ 1 & -2-\lambda & 4 \\ 1 & -1 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left. \begin{aligned} (3-\lambda)x_1 - 4x_2 + 4x_3 &= 0 \\ x_1 - (2+\lambda)x_2 + 4x_3 &= 0 \\ x_1 - x_2 + (3-\lambda)x_3 &= 0 \end{aligned} \right\} \quad \text{(I)}$$

**Case (i)** If  $\lambda = -1$ , then the equations (I) become

$$\begin{aligned} \Rightarrow &4x_1 - 4x_2 + 4x_3 = 0 \\ &x_1 - x_2 + x_3 = 0 \end{aligned}$$

$$\Rightarrow \begin{aligned} x_1 - x_2 + 4x_3 &= 0 \\ x_1 - x_2 + 4x_3 &= 0 \end{aligned}$$

The different equations are  $x_1 - x_2 + x_3 = 0$

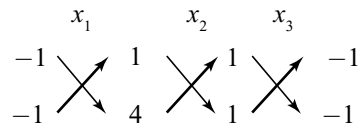
$$x_1 - x_2 + 4x_3 = 0$$

By rule of cross multiplication

$$\Rightarrow \frac{x_1}{-4+1} = \frac{x_2}{1-4} = \frac{x_3}{-1+1}$$

$$\Rightarrow \frac{x_1}{-3} = \frac{x_2}{-3} = \frac{x_3}{0}$$

$$\Rightarrow \frac{x_1}{1} = \frac{x_2}{1} = \frac{x_3}{0}$$



Choosing  $x_1 = 1, x_2 = 1, x_3 = 0$ , we get an eigen vector

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

**Case (ii)** If  $\lambda = 2$ , then equations (I) become

$$x_1 - 4x_2 + 4x_3 = 0$$

$$x_1 - 4x_2 + 4x_3 = 0$$

$$x_1 - x_2 + x_3 = 0$$

$\therefore$  the different equations are  $x_1 - 4x_2 + 4x_3 = 0$

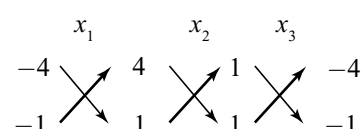
$$x_1 - x_2 + x_3 = 0$$

By the rule of cross multiplication, we get

$$\Rightarrow \frac{x_1}{-4+4} = \frac{x_2}{4-1} = \frac{x_3}{-1+4}$$

$$\Rightarrow \frac{x_1}{0} = \frac{x_2}{3} = \frac{x_3}{3}$$

$$\Rightarrow \frac{x_1}{0} = \frac{x_2}{1} = \frac{x_3}{1}$$



Choosing  $x_1 = 0, x_2 = 1, x_3 = 1$ , we get an eigen vector

$$X_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

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**Case (iii)** If  $\lambda = 3$ , then equations (I) become

$$0x_1 - 4x_2 + 4x_3 = 0 \Rightarrow 0x_1 - x_2 + x_3 = 0$$

$$x_1 - 5x_2 + 4x_3 = 0$$

$$x_1 - x_2 + 0x_3 = 0$$

The equations are different, but only two of them are independent. So, we can choose any two of them to solve. From the first two equations, we get

$$\Rightarrow \frac{x_1}{-4+5} = \frac{x_2}{1-0} = \frac{x_3}{0+1}$$

$$\Rightarrow \frac{x_1}{1} = \frac{x_2}{1} = \frac{x_3}{1}$$

$\begin{array}{ccc} x_1 & & x_2 & & x_3 \\ -1 & \nearrow & 1 & \nearrow & 0 & \nearrow & -1 \\ -5 & \searrow & 4 & \searrow & 1 & \searrow & -5 \end{array}$

Choosing  $x_1 = 1, x_2 = 1, x_3 = 1$ , we get an eigen vector  $X_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Thus the eigen values of A are  $-1, 2, 3$  and corresponding eigen vectors are

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad X_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

### Note

- (1) We are using the following integer root theorem for trial solution. “For the equation of the form

$x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x + a_0 = 0$  with integer coefficients  $a_i$ , any rational root is an integer and is a factor of the constant term  $a_0$ ”. So, it is enough we try factors of the constant term for integer solutions.

- (2) In the above problem the eigen values  $-1, 2, 3$  are different. So by property (2) the eigen vectors are linearly independent. We shall verify this:

Consider  $B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$  with the eigen vectors as rows.

$$\text{Then } |B| = 1 \cdot 0 - 1(-1) + 0 = 1 \neq 0$$

$\therefore X_1, X_2, X_3$  are linearly independent.

**Example 6**

Find the eigen values and eigen vectors of  $\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$

[AU 2004, 2007, 2010, 2012]

**Solution**

$$\text{Let } A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1$  = sum of the diagonal elements A

$$= 2 + 3 + 2 = 7$$

$S_2$  = sum of minors of the diagonal elements of determinant A

$$= \begin{vmatrix} 3 & 1 \\ 2 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ 1 & 3 \end{vmatrix}$$

$$= 6 - 2 + 4 - 1 + 6 - 2 = 4 + 3 + 4 = 11$$

$$S_3 = |A| = 2(6 - 2) - 2(2 - 1) + 1(2 - 3) = 8 - 2 - 1 = 5$$

$\therefore$  The characteristic equation is  $\lambda^3 - 7\lambda^2 + 11\lambda - 5 = 0$

Choose the integer factors of constant term  $-5$  for trial.

The integer factors of  $-5$  are  $-5, 1,$  or  $-1, 5$ .

We find  $\lambda = 1$  is a root.

Other roots are given by

$$\lambda^2 - 6\lambda + 5 = 0$$

$$\Rightarrow (\lambda - 1)(\lambda - 5) = 0$$

$$\Rightarrow \lambda = 1, 5$$

$\therefore$  the eigen values are  $\lambda = 1, 1, 5$  (Two equal eigen values)

$$\begin{array}{ccc|c} 1 & 1 & -7 & 11 & -5 \\ & 0 & 1 & -6 & 5 \\ & 1 & -6 & 5 & 0 \end{array}$$

**To find eigen vectors:**

Let  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector of A corresponding to the eigen value  $\lambda$ .

$$\text{Then } (A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

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$$\Rightarrow \left. \begin{aligned} (2-\lambda)x_1 + 2x_2 + x_3 &= 0 \\ x_1 + (3-\lambda)x_2 + x_3 &= 0 \\ x_1 + 2x_2 + (2-\lambda)x_3 &= 0 \end{aligned} \right\} \quad (I)$$

**Case (i)** If  $\lambda = 5$ , then the equations (I) become

$$-3x_1 + 2x_2 + x_3 = 0 \quad (1)$$

$$x_1 - 2x_2 + x_3 = 0 \quad (2)$$

$$x_1 + 2x_2 - 3x_3 = 0 \quad (3)$$

These 3 equations are different, but only 2 of them are independent.

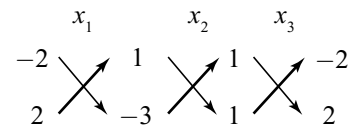
So, we can choose any two of them to solve for  $x_1, x_2, x_3$ .

From (2) and (3), by rule of cross multiplication we get

$$\Rightarrow \frac{x_1}{6-2} = \frac{x_2}{1+3} = \frac{x_3}{2+2}$$

$$\Rightarrow \frac{x_1}{4} = \frac{x_2}{4} = \frac{x_3}{4}$$

$$\Rightarrow \frac{x_1}{1} = \frac{x_2}{1} = \frac{x_3}{1}$$



Choosing  $x_1 = 1, x_2 = 1, x_3 = 1$ , we get an eigen vector

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

**Case (ii)** If  $\lambda = 1$ , then the equations (I) become

$$x_1 + 2x_2 + x_3 = 0$$

$$x_1 + 2x_2 + x_3 = 0$$

$$x_1 + 2x_2 + x_3 = 0$$

We have only one equation  $x_1 + 2x_2 + x_3 = 0$  to solve for  $x_1, x_2, x_3$ . Assign arbitrary values for two variables and solve for the third.

Choose  $x_3 = 0$ , then  $x_1 + 2x_2 = 0 \Rightarrow x_1 = -2x_2$

Choose  $x_2 = 1$ ,  $\therefore x_1 = -2$ , we get an eigen vector

$$X_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

We shall find one more solution from  $x_1 + 2x_2 + x_3 = 0$

Choose  $x_2 = 0$  then  $x_1 + x_3 = 0 \Rightarrow x_3 = -x_1$

Choose  $x_1 = 1 \quad \therefore x_3 = -1$

$\therefore$  another eigen vector corresponding to  $\lambda = 1$  is

$$X_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Thus eigen values of A are 5, 1, 1 and the corresponding eigen vectors are

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad X_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad X_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

**Note** Though the eigen values are not different, we could find independent eigen vectors.

$$\text{For, consider } B = \begin{bmatrix} 1 & 1 & 1 \\ -2 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \text{ with the vectors as rows}$$

$$\begin{aligned} \text{Then } |B| &= 1(-1-0) - 1(2-0) + 1(0-1) \\ &= -1 - 2 - 1 = -4 \neq 0 \end{aligned}$$

$\therefore X_1, X_2, X_3$  are linearly independent.

### Example 7

Find the eigen values and eigen vectors of the matrix  $\begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$

[AU 2000]

### Solution

$$\text{Let } A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$$

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 3-\lambda & 10 & 5 \\ -2 & -3-\lambda & -4 \\ 3 & 5 & 7-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1$  = sum of the diagonal elements of A

$$= 3 + (-3) + 7 = 7$$

$S_2$  = sum of minors of elements of the diagonal of A

$$\begin{aligned}
 &= \begin{vmatrix} -3 & -4 \\ 5 & 7 \end{vmatrix} + \begin{vmatrix} 3 & 5 \\ 3 & 7 \end{vmatrix} + \begin{vmatrix} 3 & 10 \\ -2 & -3 \end{vmatrix} \\
 &= (-21 + 20) + (21 - 15) + (-9 + 20) \\
 &= -1 + 6 + 11 = 16
 \end{aligned}$$

$$\begin{aligned}
 S_3 &= |A| = 3(-21 + 20) - 10(-14 + 12) + 5(-10 + 9) \\
 &= -3 + 20 - 5 = 12
 \end{aligned}$$

∴ the characteristic equation is  $\lambda^3 - 7\lambda^2 + 16\lambda - 12 = 0$

Choose the integer factors of constant term  $-12$  for trial.

We find  $\lambda = 2$  is a root.

Other roots are given by

$$\lambda^2 - 5\lambda + 6 = 0$$

$$\Rightarrow (\lambda - 2)(\lambda - 3) = 0 \Rightarrow \lambda = 2, 3$$

$$\begin{array}{r|rrrr}
 2 & 1 & -7 & 16 & -12 \\
 & 0 & 2 & -10 & 12 \\
 \hline
 & 1 & -5 & 6 & 0
 \end{array}$$

∴ The eigen values are  $\lambda = 2, 2, 3$ . (Two equal eigen values)

**To find eigen vectors:**

Let  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector corresponding to the eigen value  $\lambda$ .

$$\text{Then} \quad (A - \lambda I)X = 0$$

$$\Rightarrow \left. \begin{aligned} (3 - \lambda)x_1 + 10x_2 + 5x_3 &= 0 \\ -2x_1 - (3 + \lambda)x_2 - 4x_3 &= 0 \\ 3x_1 + 5x_2 + (7 - \lambda)x_3 &= 0 \end{aligned} \right\} \quad (I)$$

**Case (i)** If  $\lambda = 3$ , then the equations (I) become

$$0x_1 + 10x_2 + 5x_3 = 0 \Rightarrow 0x_1 + 2x_2 + x_3 = 0 \quad (1)$$

$$-2x_1 - 6x_2 - 4x_3 = 0 \Rightarrow x_1 + 3x_2 + 2x_3 = 0 \quad (2)$$

$$3x_1 + 5x_2 + 4x_3 = 0 \quad (3)$$

These 3 equations are different, but only 2 of them are independent. From (1) and (2), by rule of cross multiplication we get

$$\begin{aligned}
 \Rightarrow \quad \frac{x_1}{1} &= \frac{x_2}{1} = \frac{x_3}{-2} \\
 &\quad \begin{array}{ccc} x_1 & x_2 & x_3 \\ 2 \swarrow \searrow & 1 \swarrow \searrow & 0 \swarrow \searrow \\ 3 & 2 & 1 \end{array}
 \end{aligned}$$



Choosing  $x_1 = 1, x_2 = 1, x_3 = -2$ , we get an eigen vector

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

**Case (ii)** If  $\lambda = 2$ , then equations (I) become

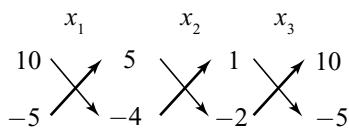
$$x_1 + 10x_2 + 5x_3 = 0 \quad (1)$$

$$-2x_1 - 5x_2 - 4x_3 = 0 \quad (2)$$

$$3x_1 + 5x_2 + 5x_3 = 0 \quad (3)$$

From (1) and (2), by rule of cross multiplication we get

$$\begin{aligned} \Rightarrow \quad \frac{x_1}{-40+25} &= \frac{x_2}{-10+4} = \frac{x_3}{-5+20} \\ \Rightarrow \quad \frac{x_1}{-15} &= \frac{x_2}{-6} = \frac{x_3}{15} \end{aligned}$$



$$\Rightarrow \quad \frac{x_1}{5} = \frac{x_2}{2} = \frac{x_3}{-5}$$

Choosing  $x_1 = 5, x_2 = 2, x_3 = -5$ , we get an eigen vector  $X_2 = \begin{bmatrix} 5 \\ 2 \\ -5 \end{bmatrix}$

In this case we cannot choose another vector independent of  $X_2$ . So corresponding to the equal eigen values 2, 2 the eigen vectors are equal  $X_2 = \begin{bmatrix} 5 \\ 2 \\ -5 \end{bmatrix} = X_3$

Thus the eigen values of A are 3, 2, 2 and the corresponding eigen vectors are

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \quad X_2 = X_3 = \begin{bmatrix} 5 \\ 2 \\ -5 \end{bmatrix}$$

**Note** Here  $X_1$  and  $X_2$  are linearly independent vectors as they belong to different eigen values, but  $X_1, X_2, X_3$  are linearly dependent vectors. See note (3) below properties. (Page 1.6)

### Example 8

Find the eigen values and eigen vectors of the matrix  $\begin{bmatrix} 6 & -6 & 5 \\ 14 & -13 & 10 \\ 7 & -6 & 4 \end{bmatrix}$

[MKU 1998]

**Solution**

$$\text{Let } A = \begin{bmatrix} 6 & -6 & 5 \\ 14 & -13 & 10 \\ 7 & -6 & 4 \end{bmatrix}$$

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 6-\lambda & -6 & 5 \\ 14 & -13-\lambda & 10 \\ 7 & -6 & 4-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1$  = sum of the diagonal elements of A

$$= 6 + (-13) + 4 = -3$$

$S_2$  = sum of minors of elements of the diagonal of A

$$= \begin{vmatrix} -13 & 10 \\ -6 & 4 \end{vmatrix} + \begin{vmatrix} 6 & 5 \\ 7 & 4 \end{vmatrix} + \begin{vmatrix} 6 & -6 \\ 14 & -13 \end{vmatrix}$$

$$= (-52 + 60) + (24 - 35) + (-78 + 84) = 8 - 11 + 6 = 3$$

$$S_3 = |A| = 6(-52 + 60) + 6(56 - 70) + 5(-84 + 91)$$

$$= 48 + 6(-14) + 5(7) = 48 - 84 + 35 = -1$$

$\therefore$  the characteristic equation is  $\lambda^3 + 3\lambda^2 + 3\lambda + 1 = 0$

$$\Rightarrow (\lambda + 1)^3 = 0$$

$$\Rightarrow \lambda = -1, -1, -1$$

(3 equal eigen values)

**To find eigen vectors:**

Let  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector corresponding to the eigen value  $\lambda$ .

$$\text{Then } (A - \lambda I)X = 0$$

$$\Rightarrow \begin{bmatrix} 6-\lambda & -6 & 5 \\ 14 & -13-\lambda & 10 \\ 7 & -6 & 4-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left. \begin{aligned} (6-\lambda)x_1 - 6x_2 + 5x_3 &= 0 \\ 14x_1 - (13+\lambda)x_2 + 10x_3 &= 0 \\ 7x_1 - 6x_2 + (4-\lambda)x_3 &= 0 \end{aligned} \right\} \quad \text{(I)}$$

If  $\lambda = -1$ , then the equations (I) become

$$\begin{aligned}
7x_1 - 6x_2 + 5x_3 &= 0 \\
14x_1 - 12x_2 + 10x_3 &= 0 \Rightarrow 7x_1 - 6x_2 + 5x_3 = 0 \\
7x_1 - 6x_2 + 5x_3 &= 0
\end{aligned}$$

We have only one equation  $7x_1 - 6x_2 + 5x_3 = 0$

Assign arbitrary values to two variables and find the third.

We shall find 3 vectors.

Putting  $x_1 = 0$ , we get  $-6x_2 + 5x_3 = 0$

$$\Rightarrow 6x_2 = 5x_3 \Rightarrow \frac{x_2}{5} = \frac{x_3}{6}$$

Choosing  $x_2 = 5$ ,  $x_3 = 6$ , we get an eigen vector  $X_1 = \begin{bmatrix} 0 \\ 5 \\ 6 \end{bmatrix}$

Putting  $x_2 = 0$ , we get  $7x_1 + 5x_3 = 0$

$$\Rightarrow 7x_1 = -5x_3 \Rightarrow \frac{x_1}{5} = \frac{x_3}{-7}$$

Choosing  $x_1 = 5$ ,  $x_3 = -7$ , we get the second eigen vector  $X_2 = \begin{bmatrix} 5 \\ 0 \\ -7 \end{bmatrix}$

Again, putting  $x_3 = 0$ , we get  $7x_1 - 6x_2 = 0 \Rightarrow 7x_1 = 6x_2$

$$\Rightarrow \frac{x_1}{6} = \frac{x_2}{7}$$

Choosing  $x_1 = 6$ ,  $x_2 = 7$ , we get the third eigen vector  $X_3 = \begin{bmatrix} 6 \\ 7 \\ 0 \end{bmatrix}$

Thus eigen values of A are  $-1, -1, -1$  and the corresponding eigen vectors are

$X_1 = \begin{bmatrix} 0 \\ 5 \\ 6 \end{bmatrix}$ ,  $X_2 = \begin{bmatrix} 5 \\ 0 \\ -7 \end{bmatrix}$  and  $X_3 = \begin{bmatrix} 6 \\ 7 \\ 0 \end{bmatrix}$ , which are different

**Note** If  $B = \begin{bmatrix} 0 & 5 & 6 \\ 5 & 0 & 7 \\ 6 & -7 & 0 \end{bmatrix}$  with the eigen vectors as rows,

$$\text{then } |B| = 0 - 5(0 - 42) + 6(-35 - 0) = 210 - 210 = 0$$

$\therefore$  The vectors  $X_1, X_2, X_3$  are linearly dependent. However, any two of them are linearly independent. Geometrically, it means that all the vectors are coplanar, but any two of them are non-collinear.

In this example we have seen  $-1$  is the only eigen value of  $3 \times 3$  matrix and two linearly independent eigen vectors.

**Example 9**

Find the eigen values and eigen vectors of  $\begin{bmatrix} -5 & -5 & -9 \\ 8 & 9 & 18 \\ -2 & -3 & -7 \end{bmatrix}$

**Solution**

$$\text{Let } A = \begin{bmatrix} -5 & -5 & -9 \\ 8 & 9 & 18 \\ -2 & -3 & -7 \end{bmatrix}$$

The characteristic equation is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} -5-\lambda & -5 & -9 \\ 8 & 9-\lambda & 18 \\ -2 & -3 & -7-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$\text{where } S_1 = -5 + 9 + (-7) = -3$$

$$S_2 = \begin{vmatrix} 9 & 18 \\ -3 & -7 \end{vmatrix} + \begin{vmatrix} -5 & -9 \\ -2 & -7 \end{vmatrix} + \begin{vmatrix} -5 & -5 \\ 8 & 9 \end{vmatrix}$$

$$= (-63 + 54) + (35 - 18) + (-45 + 40) = -9 + 17 - 5 = 3$$

$$S_3 = |A| = -5(-63 + 54) + 5(-56 + 36) - 9(-24 + 18)$$

$$= 45 - 100 + 54 = -1$$

$\therefore$  the characteristic equation is  $\lambda^3 + 3\lambda^2 + 3\lambda + 1 = 0$

$$\Rightarrow (\lambda + 1)^3 = 0$$

$$\Rightarrow \lambda = -1, -1, -1$$

If  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  is an eigen vector of the eigen value  $\lambda$ ,

$$\text{then } (A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} -5-\lambda & -5 & -9 \\ 8 & 9-\lambda & 18 \\ -2 & -3 & -7-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} (-5-\lambda)x_1 - 5x_2 - 9x_3 &= 0 \\ 8x_1 + (9-\lambda)x_2 + 18x_3 &= 0 \\ -2x_1 - 3x_2 - (7+\lambda)x_3 &= 0 \end{aligned}$$

If  $\lambda = -1$ , then these equations become

$$-4x_1 - 5x_2 - 9x_3 = 0 \Rightarrow 4x_1 + 5x_2 + 9x_3 = 0$$

$$8x_1 + 10x_2 + 18x_3 = 0 \Rightarrow 4x_1 + 5x_2 + 9x_3 = 0$$

$$-2x_1 - 3x_2 - 6x_3 = 0 \Rightarrow 2x_1 + 3x_2 + 6x_3 = 0$$

From the last two equations, by rule of cross multiplication, we get

$$\Rightarrow \frac{x_1}{30-27} = \frac{x_2}{18-24} = \frac{x_3}{12-10}$$

$$\Rightarrow \frac{x_1}{3} = \frac{x_2}{-6} = \frac{x_3}{2}$$

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ 5 & 9 & 4 \\ 3 & 6 & 2 \end{array} \begin{array}{c} \nearrow \searrow \\ \nearrow \searrow \\ \nearrow \searrow \end{array} \begin{array}{ccc} 9 & 4 & 5 \\ 6 & 2 & 3 \end{array}$$

Choosing  $x_1 = 3$ ,  $x_2 = -6$ ,  $x_3 = 2$ , we get only one independent eigen vector

$$X_1 = \begin{bmatrix} 3 \\ -6 \\ 2 \end{bmatrix}. \text{ So we take } X_1 = X_2 = X_3 = \begin{bmatrix} 3 \\ -6 \\ 2 \end{bmatrix}$$

**Note** In this example,  $3 \times 3$  matrix has one eigen value  $-1$  and one linearly independent eigen vector.

### Example 10

Find the eigen values and eigen vectors of  $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$  [AU 2009]

### Solution

$$\text{Given } A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1$  = sum of diagonal elements of A

$$= 2 + 2 + 1 = 5$$

$S_2$  = sum of minors of elements of the diagonals of A

$$= \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix}$$

$$= 2 - 0 + 2 - 0 + 4 - 1 = 7$$

$$S_3 = |A| = 2(2-0) - 1(1-0) + 1(0) = 3$$

$\therefore$  the characteristic equation is  $\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$

Since the sum of the coefficients is zero,  $\lambda = 1$  is a root.

The other roots are given by

$$\lambda^2 - 4\lambda + 3 = 0$$

$$\Rightarrow (\lambda - 1)(\lambda - 3) = 0 \Rightarrow \lambda = 1, 3$$

$\therefore$  The eigen values are  $\lambda = 1, 1, 3$

$$\begin{array}{c|cccc} 1 & 1 & -5 & 7 & -3 \\ & 0 & 1 & -4 & 3 \\ \hline & 1 & -4 & 3 & 0 \end{array}$$

**To find eigen vectors:**

Let  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector corresponding to an eigen value  $\lambda$  of A.

$$\text{Then } (A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} 2 - \lambda & 1 & 1 \\ 1 & 2 - \lambda & 1 \\ 0 & 0 & 1 - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left. \begin{aligned} (2 - \lambda)x_1 + x_2 + x_3 &= 0 \\ x_1 + (2 - \lambda)x_2 + x_3 &= 0 \\ 0x_1 + 0x_2 + (1 - \lambda)x_3 &= 0 \end{aligned} \right\} \quad \text{(I)}$$

**Case (i)** If  $\lambda = 1$ , then equations (I) become

$$x_1 + x_2 + x_3 = 0$$

$$x_1 + x_2 + x_3 = 0$$

$$0x_3 = 0$$

Thus we have  $x_1 + x_2 + x_3 = 0$

Put  $x_1 = 1, x_2 = 0$ , then  $x_3 = -1$

$$\therefore \text{ an eigen vector is } X_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Since  $\lambda = 1$  is a repeated eigen value, we have to find another eigen vector. Put  $x_1 = 0, x_2 = 1 \therefore x_3 = -1$

$$\text{So, another eigen vector is } X_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

**Case (ii)** If  $\lambda = 3$  then equations (I) become

$$-x_1 + x_2 + x_3 = 0$$

$$\Rightarrow x_1 - x_2 - x_3 = 0$$

$$\text{and } -2x_3 = 0 \Rightarrow x_3 = 0$$

$$\therefore x_1 - x_2 = 0 \Rightarrow x_1 = x_2$$

Take  $x_2 = 1 \therefore x_1 = 1, x_2 = 1, x_3 = 0$

$\therefore$  an eigen vector is  $X_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

Thus the eigen values are  $\lambda = 1, 1, 3$  and the eigen vectors are

$$X_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \quad X_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

## Properties of eigen values

**1. A square matrix  $A$  and its transpose  $A^T$  have the same eigen values.**

**Proof** Eigen values of  $A$  are the roots of its characteristic equation

$$|A - \lambda I| = 0 \quad (1)$$

$$\text{We know } (A - \lambda I)^T = A^T - (\lambda I)^T \quad [\because (A + B)^T = A^T + B^T]$$

$$= A^T - \lambda I^T$$

$$= A^T - \lambda I \quad [\because I^T = I]$$

$$\therefore |(A - \lambda I)^T| = |A^T - \lambda I| \quad (2)$$

$$\text{For any square matrix } B, |B^T| = |B|$$

$$\therefore |(A - \lambda I)^T| = |A - \lambda I| \quad (3)$$

$$\text{From (2) and (3), } |A - \lambda I| = |A^T - \lambda I|$$

This shows that the characteristic polynomial of  $A$  and  $A^T$  are the same. Hence the characteristic equations of  $A$  and  $A^T$  is (1).

$\therefore A$  and  $A^T$  have the same eigen values. ■

**2. Sum of the eigen values of a square matrix  $A$  is equal to the sum of the elements on its main diagonal.**

**Proof** Let  $A$  be a square matrix of order  $n$ .

Then the characteristic equation of  $A$  is  $|A - \lambda I| = 0$

$$\Rightarrow \lambda^n - S_1 \lambda^{n-1} + S_2 \lambda^{n-2} - \dots + (-1)^n S_n = 0 \quad (1)$$

where  $S_1$  = sum of the diagonal elements of  $A$

If  $\lambda_1, \lambda_2, \dots, \lambda_n$  are the roots of (1), then  $\lambda_1, \lambda_2, \dots, \lambda_n$  are the eigen values of  $A$ .

From theory of equations, sum of the roots of (1) is

$$= \frac{-\text{coefficient of } \lambda^{n-1}}{\text{coefficient of } \lambda^n}$$

$\Rightarrow \lambda_1 + \lambda_2 + \dots + \lambda_n = -(-S_1) = S_1$   
 $\therefore$  the sum of the eigen values  $= \lambda_1 + \lambda_2 + \dots + \lambda_n = S_1$   
 $=$  sum of the diagonal elements of the matrix A. ■

**Note** Sum of the diagonal elements of A is called **the trace of A**.

$\therefore$  **Sum of the eigen values = trace of A**

**3. Product of the eigen values of a square matrix A is equal to  $|A|$ .**

**Proof** Let A be a square matrix of order  $n$ .

Then its characteristic equation is  $|A - \lambda I| = 0$

$$\Rightarrow \lambda^n - S_1 \lambda^{n-1} + S_2 \lambda^{n-2} - \dots + (-1)^n S_n = 0 \quad (1)$$

where  $S_n = |A|$ .

If  $\lambda_1, \lambda_2, \dots, \lambda_n$  are the  $n$  roots of (1), then from theory of equations,  
the product of roots  $= (-1)^n \frac{\text{constant term}}{\text{coefficient of } \lambda^n}$

$$\begin{aligned} \Rightarrow \lambda_1 \lambda_2 \dots \lambda_n &= (-1)^n (-1)^n S_n \\ &= (-1)^{2n} S_n \\ &= S_n = |A| \quad [\because (-1)^{2n} = 1] \end{aligned}$$

$\therefore$  Product of the eigen values  $= \lambda_1 \lambda_2 \dots \lambda_n = S_n = |A|$ . ■

**Note** If at least one eigen value is 0, then  $|A| = 0$

$\therefore$  A is a singular matrix. If all the eigen values are non-zero, then  $|A| \neq 0$

$\therefore$  A is a non-singular if all the eigen values are non-zero.

**4. If  $\lambda_1, \lambda_2, \dots, \lambda_n$  are non-zero eigen values of square matrix of order  $n$ ,**

**then  $\frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}$  are eigen values of  $A^{-1}$ . [AU 2010]**

**Proof** Let  $\lambda$  be any non-zero eigen value of A, then there exists a non-zero column matrix X such that  $AX = \lambda X$ . Since all the eigen values are non-zero, A is non-singular.

$\therefore A^{-1}$  exists.



$$\begin{aligned}
\therefore \quad & A^{-1} (AX) = A^{-1} (\lambda X) \\
\Rightarrow & (A^{-1} A)X = \lambda(A^{-1} X) \\
\Rightarrow & IX = \lambda(A^{-1} X) \\
\Rightarrow & X = \lambda(A^{-1} X)
\end{aligned}$$

$$\Rightarrow \quad \frac{1}{\lambda} X = A^{-1} X \quad [\because \lambda \neq 0]$$

$$\Rightarrow \quad A^{-1} X = \frac{1}{\lambda} X.$$

So  $\frac{1}{\lambda}$  is an eigen value of  $A^{-1}$ .

This is true for all the eigen values of A.

$\therefore \frac{1}{\lambda_1}, \frac{1}{\lambda_2}, \dots, \frac{1}{\lambda_n}$  are the eigen values of  $A^{-1}$ . ■

Note that the eigen vector for  $A^{-1}$  corresponding to  $\frac{1}{\lambda}$  is also X.

**5. If  $\lambda_1, \lambda_2, \dots, \lambda_n$  are the eigen values of A, then**

(i)  $c\lambda_1, c\lambda_2, \dots, c\lambda_n$  are the eigen values of  $cA$ , where  $c \neq 0$

(ii)  $\lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$  are the eigen values of  $A^m$ , where m is a positive integer.

**Proof** Let  $\lambda$  be any eigen value of A, then there exists a non-zero column matrix X such that

$$AX = \lambda X \quad (1)$$

(i) Multiply by  $c \neq 0$  then  $c(AX) = c(\lambda X)$

$$\Rightarrow (cA) X = (c\lambda) X$$

$\therefore c\lambda$  is an eigen value of  $cA$ . This is true for all eigen values of A.

$\therefore c\lambda_1, c\lambda_2, \dots, c\lambda_n$  are the eigen values of  $cA$ .

(ii) Now  $A^2X = A(AX) = A(\lambda X)$  [using (1)]

$$= \lambda(AX)$$

$$= \lambda(\lambda X)$$

$\therefore A^2X = \lambda^2X \Rightarrow \lambda^2$  is an eigen value of  $A^2$ .

Similarly  $A^3X = A(A^2X)$

$$= A(\lambda^2X)$$

$$= \lambda^2(AX)$$

$$= \lambda^2(\lambda X)$$

$A^3X = \lambda^3X \Rightarrow \lambda^3$  is an eigen value of  $A^3$ .

Proceeding in this way, we have  $A^m X = \lambda^m X$  for any positive integer  $m$ . This is true for all eigen values.

$\therefore \lambda_1^m, \lambda_2^m, \dots, \lambda_n^m$  are eigen values of  $A^m$ . ■

**6. If  $\lambda_1, \lambda_2, \dots, \lambda_n$  are the eigen values of  $A$ , then**

**(i)  $\lambda_1 - k, \lambda_2 - k, \dots, \lambda_n - k$  are the eigen values of  $A - KI$ .**

**(ii)  $\alpha_0 \lambda_1^2 + \alpha_1 \lambda_1 + \alpha_2, \alpha_0 \lambda_2^2 + \alpha_1 \lambda_2 + \alpha_2, \dots, \alpha_0 \lambda_n^2 + \alpha_1 \lambda_n + \alpha_2$  are the eigen values of  $\alpha_0 A^2 + \alpha_1 A + \alpha_2 I$ .**

**Proof** Let  $\lambda$  be any eigen value of  $A$

$$\text{then } AX = \lambda X \quad (1)$$

where  $X \neq 0$  is a column matrix.

$$\therefore AX - KX = \lambda X - KX$$

$$\Rightarrow (A - KI)X = (\lambda - K)X$$

$\therefore \lambda - K$  is an eigen value of  $A - KI$ . This is true for all eigen values of  $A$ .

$\therefore \lambda_1 - K, \lambda_2 - K, \dots, \lambda_n - K$  are the eigen values of  $A - KI$ .

(ii) We have  $AX = \lambda X$  and  $A^2 X = \lambda^2 X$ .

$$\therefore \alpha_0 (A^2 X) = \alpha_0 (\lambda^2 X)$$

$$\text{and } \alpha_1 (AX) = \alpha_1 (\lambda X)$$

$$\therefore \alpha_0 (A^2 X) + \alpha_1 (AX) = \alpha_0 (\lambda^2 X) + \alpha_1 (\lambda X)$$

Adding  $\alpha_2 X$  on both sides, we get

$$\alpha_0 (A^2 X) + \alpha_1 (AX) + \alpha_2 X = \alpha_0 (\lambda^2 X) + \alpha_1 (\lambda X) + \alpha_2 X$$

$$\Rightarrow (\alpha_0 A^2 + \alpha_1 A + \alpha_2 I)X = (\alpha_0 \lambda^2 + \alpha_1 \lambda + \alpha_2)X$$

This means  $\alpha_0 \lambda^2 + \alpha_1 \lambda + \alpha_2$  is an eigen value of  $\alpha_0 A^2 + \alpha_1 A + \alpha_2 I$ .

This is true for all eigen values of  $A$ .

$\therefore \alpha_0 \lambda_1^2 + \alpha_1 \lambda_1 + \alpha_2, \alpha_0 \lambda_2^2 + \alpha_1 \lambda_2 + \alpha_2, \dots, \alpha_0 \lambda_n^2 + \alpha_1 \lambda_n + \alpha_2$  are the eigen values of  $\alpha_0 A^2 + \alpha_1 A + \alpha_2 I$ . ■

### Note

1. The eigen values of the unit matrix  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$  are 1, 1, 1 and the

corresponding eigen vectors are  $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ , which are independent.

2. The eigen values of a triangular matrix  $\begin{bmatrix} \lambda_1 & a_{12} & a_{13} \\ 0 & \lambda_2 & a_{23} \\ 0 & 0 & \lambda_3 \end{bmatrix}$  are the main diagonal elements  $\lambda_1, \lambda_2, \lambda_3$ .
3. If  $\lambda$  is an eigen value of  $A$  then  $AX = \lambda X$ . We have seen  $A^2X = \lambda^2X, \dots, A^mX = \lambda^mX$ .

Thus the eigen values of  $A, A^2, \dots, A^m$  are  $\lambda, \lambda^2, \dots, \lambda^m$  which are all different. But they all have the same eigen vector  $X$ .

Similarly,  $\lambda$  and  $\alpha_0\lambda^2 + \alpha_1\lambda + \alpha_2$  are eigen values of  $A$  and  $\alpha_0A^2 + \alpha_1A + \alpha_2I$ . But they have the same eigen vector  $X$ .

## WORKED EXAMPLES

### Example 11

Find the sum and product of the eigen values of the matrix

$$\begin{bmatrix} 1 & 2 & -2 \\ 1 & 0 & 3 \\ -2 & -1 & -3 \end{bmatrix}$$

#### Solution

Sum of the eigen values = Sum of the elements on the main diagonal

$$= 1 + 0 + (-3) = -2$$

$$\begin{aligned} \text{Product of the eigen values} = |A| &= \begin{vmatrix} 1 & 2 & -2 \\ 1 & 0 & 3 \\ -2 & -1 & -3 \end{vmatrix} \\ &= 1(0 + 3) - 2(-3 + 6) - 2(-1 - 0) \\ &= 3 - 6 + 2 = -1 \end{aligned}$$

### Example 12

If 2 and 3 are eigen values of  $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$ , find the eigen values of  $A^{-1}$  and  $A^3$ . [AU 2000]

#### Solution

Given 2 and 3 are two eigen values of  $A$ .

Let  $\lambda$  be the 3<sup>rd</sup> eigen value.

We know, sum of the eigen values = sum of the diagonal elements.

$$\Rightarrow 2 + 3 + \lambda = 3 + (-3) + 7$$

$$\Rightarrow \lambda = 2$$

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So, eigen values of A are 2, 2, 3

∴ Eigen values of  $A^{-1}$  are  $\frac{1}{2}, \frac{1}{2}, \frac{1}{3}$  and eigen values of  $A^3$  are  $2^3, 2^3, 3^3 \Rightarrow 8, 8, 27$ .

**Example 13**

If  $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$  is an eigen vector of the matrix  $\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ , find the corresponding eigen value.

**Solution**

Let  $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$  and  $X = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ . If  $\lambda$  is the eigen value,

then  $(A - \lambda I)X = 0$

$$\Rightarrow \begin{bmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow (5-\lambda)4 + 4 = 0 \Rightarrow \lambda = 6$$

**Example 14**

If  $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$  find the eigen values of  $A^2 - 2A + I$ .

**Solution**

Characteristic equations of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 3-\lambda & 1 & 4 \\ 0 & 2-\lambda & 6 \\ 0 & 0 & 5-\lambda \end{vmatrix} = 0$$

Expanding by  $C_1$ ,  $(3 - \lambda)(2 - \lambda)(5 - \lambda) = 0$

⇒  $\lambda = 3, 2, 5$  are the eigen values of A.

∴ the eigen values of  $A^2 - 2A + I$  are

$$3^2 - 2 \cdot 3 + 1, 2^2 - 2 \cdot 2 + 1, 5^2 - 2 \cdot 5 + 1$$

i.e., the eigen values of  $A^2 - 2A + I$  are 4, 1, 16.

**Example 15**

The product of two eigen values of the matrix  $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$  is 16.

Find the third eigen value.

[AU 2003, 2012]

**Solution**

Let  $\lambda_1, \lambda_2, \lambda_3$  be the eigen values of A.

Given  $\lambda_1 \cdot \lambda_2 = 16$

We know that  $\lambda_1 \cdot \lambda_2 \cdot \lambda_3 = |A|$

$$\begin{aligned} \Rightarrow 16\lambda_3 &= \begin{vmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{vmatrix} \\ &= 6(9-1) + 2(-6+2) + 2(2-6) \\ \Rightarrow 16\lambda_3 &= 48 - 8 - 8 = 32 \Rightarrow \lambda_3 = 2 \end{aligned}$$

**Example 16**

Find the eigen values of the matrix  $\begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$ . Hence find the matrix whose eigen values are  $\frac{1}{6}$  and  $-1$ . [AU 2001]

**Solution**

Let  $A = \begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$ .

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\begin{aligned} \Rightarrow & \begin{vmatrix} 1-\lambda & -2 \\ -5 & 4-\lambda \end{vmatrix} = 0 \\ \Rightarrow & \lambda^2 - S_1\lambda + S_2 = 0 \end{aligned}$$

where  $S_1 = 1 + 4 = 5$  and  $S_2 = |A| = 4 - 10 = -6$

$\therefore$  the characteristic equation is  $\lambda^2 - 5\lambda - 6 = 0$

$$\Rightarrow (\lambda - 6)(\lambda + 1) = 0 \Rightarrow \lambda = 6, -1$$

Since 6, -1 are the eigen values of A, by property (4),  $\frac{1}{6}, -1$  are the eigen values of  $A^{-1}$ .

$$\begin{aligned} \text{So, the required matrix is } A^{-1} &= \frac{1}{|A|} \text{adj } A = -\frac{1}{6} \begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}^T \\ &= -\frac{1}{6} \begin{bmatrix} 4 & 2 \\ 5 & 1 \end{bmatrix} \end{aligned}$$

**Example 17**

If  $\alpha, \beta$  are the eigen values of  $\begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix}$  form the matrix whose eigen values are  $\alpha^3, \beta^3$ . [AU 2001]

**Solution**

Let  $A = \begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix}$ . Since  $\alpha, \beta$  are the eigen values of  $A$ , by property 5(ii),

$\alpha^3, \beta^3$  are eigen values of  $A^3$ .

$$\text{Now } A^2 = A \cdot A = \begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix} = \begin{bmatrix} 10 & -8 \\ -8 & 26 \end{bmatrix}$$

$$\text{and } A^3 = A^2 A = \begin{bmatrix} 10 & -8 \\ -8 & 26 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix} = \begin{bmatrix} 38 & -50 \\ -50 & 138 \end{bmatrix}$$

**EXERCISE 1.1**

Find eigen values and eigen vectors of the following matrices.

$$1. \begin{bmatrix} 2 & 2 & 0 \\ 2 & 1 & 1 \\ -7 & 2 & -3 \end{bmatrix}$$

[AU 1998]

$$2. \begin{bmatrix} 4 & -20 & -10 \\ -2 & 10 & 4 \\ 6 & -30 & -13 \end{bmatrix}$$

[AU 1999]

$$3. \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

[AU 1996, 2009, 2010]

$$4. \begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$$

[AU 2001]

$$5. \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

[MU 1998, AU 2008, 2009]

$$6. \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

[AU 2010]

$$7. \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

[MKU 1998]

**ANSWERS TO EXERCISE 1.1**

$$1. \lambda = 1, 3, -4; \text{ eigen vectors } \begin{bmatrix} 2 \\ -1 \\ -4 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ 13 \end{bmatrix}$$

$$2. \lambda = 0, -1, 2; \text{ eigen vectors } \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

3.  $\lambda = -3, -3, 5$ ; eigen vectors  $\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$

4.  $\lambda = 3, 6, 9$ ; eigen vectors  $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$

5.  $\lambda = 2, 2, 8$ ; eigen vectors  $\begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

6.  $\lambda = -2, 3, 6$ ; eigen vectors  $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

7.  $\lambda = 1, 1, 4$ ; eigen vectors  $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

### 1.3 CAYLEY-HAMILTON THEOREM

#### Theorem 1.3.1 Every square matrix satisfies its characteristic equation

Cayley-Hamilton theorem has two important uses (1) to find the inverse of a non-singular matrix  $A$  and (2) to find higher integral powers of  $A$ .

#### WORKED EXAMPLES

##### Example 1

Verify that  $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$  satisfies its own characteristic equation and hence find  $A^4$ . [AU 2003]

##### Solution

Given  $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$

The characteristic equation of  $A$  is  $|A - \lambda I| = 0$

$$\Rightarrow \lambda^2 - S_1\lambda + S_2 = 0$$

where  $S_1 = 1 + (-1) = 0$ ,  $S_2 = |A| = -1 - 4 = -5$

$\therefore$  the characteristic equation is  $\lambda^2 - 5 = 0$  (1)

By Cayley-Hamilton theorem,  $A$  satisfies (1)

i.e.  $A^2 - 5I = 0$  (2)

We shall now verify this by direct computation.

$$A^2 = A \cdot A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{aligned} \therefore A^2 - 5I &= \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

$$\Rightarrow A^2 - 5I = 0.$$

Hence A satisfies its characteristic equation.

**To find  $A^4$ :** We have  $A^2 = 5I$

[from (2)]

$$\begin{aligned} \therefore A^4 &= 5A^2 \\ &= 5 \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 25 & 0 \\ 0 & 25 \end{bmatrix} \end{aligned}$$

### Example 2

**Verify Cayley-Hamilton theorem for the matrix  $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$  and find its inverse. Also express  $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$  as a linear polynomial in A.**

### Solution

$$\text{Given } A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \lambda^2 - S_1\lambda + S_2 = 0$$

where  $S_1 = 1 + 3 = 4$ ,  $S_2 = |A| = 3 - 8 = -5$

$$\therefore \text{ the characteristic equation is } \lambda^2 - 4\lambda - 5 = 0 \quad (1)$$

By Cayley-Hamilton theorem, A satisfies (1)

$$\therefore A^2 - 4A - 5I = 0 \quad (2)$$

We shall now verify this by direct computations.

$$\text{Now } A^2 = A \cdot A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix}$$

$$A^2 - 4A - 5I = \begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix} - 4 \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$\begin{aligned}
&= \begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix} - \begin{bmatrix} 4 & 16 \\ 8 & 13 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \\
&= \begin{bmatrix} 9-4-5 & 16-16-0 \\ 8-8-0 & 17-12-5 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}
\end{aligned}$$

$\Rightarrow A^2 - 4A - 5I = 0$ . Hence the theorem is verified.

**To find  $A^{-1}$ :** We have

$$5I = A^2 - 4A$$

Multiply by  $A^{-1}$ , we get  $5A^{-1} = A^{-1}A^2 - 4A^{-1}A$

$$= A - 4I$$

$$= \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 1-4 & 4-0 \\ 2-0 & 3-4 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{5} \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

Finally, to find  $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ :

Consider the polynomial  $\lambda^5 - 4\lambda^4 - 7\lambda^3 + 11\lambda^2 - \lambda - 10$

Divide it by  $\lambda^2 - 4\lambda - 5$ . We get the quotient  $\lambda^3 - 2\lambda + 3$  and remainder  $\lambda + 5$

$$\therefore \lambda^5 - 4\lambda^4 - 7\lambda^3 + 11\lambda^2 - \lambda - 10 = (\lambda^2 - 4\lambda - 5)(\lambda^3 - 2\lambda + 3) + \lambda + 5$$

Replace  $\lambda$  by  $A$ , we get

$$\begin{aligned}
A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I &= (A^2 - 4A - 5I)(A^3 - 2A + 3) + A + 5I \\
&= 0 + A + 5I && \text{[using (2)]} \\
&= A + 5I,
\end{aligned}$$

which is a linear polynomial in  $A$ .

### Example 3

Find the characteristic equation of the matrix  $A$  given

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}. \text{ Hence find } A^{-1} \text{ and } A^4.$$

[AU 2009, 2010]

**Solution**

$$\text{Given } A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1$  = sum of the diagonal elements of A

$$= 2 + 2 + 2 = 6$$

$S_2$  = sum of minors of the diagonal elements of A

$$= \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix}$$

$$= 4 - 1 + 4 - 1 + 4 - 1 = 9$$

$$S_3 = |A| = 2(4 - 1) + (-2 + 1) + (1 - 2) = 6 - 1 - 1 = 4$$

$\therefore$  the characteristic equation is  $\lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$

By Cayley-Hamilton theorem, A satisfies its characteristic equation

$$\therefore A^3 - 6A^2 + 9A - 4I = 0 \quad (1)$$

$$\Rightarrow 4I = A^3 - 6A^2 + 9A$$

$$\text{Multiply by } A^{-1}, \quad 4IA^{-1} = A^3A^{-1} - 6A^2 \cdot A^{-1} + 9AA^{-1}$$

$$\Rightarrow 4A^{-1} = A^2 - 6A + 9I$$

$$\text{But } A^2 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 4+1+1 & -2-2-1 & 2+1+2 \\ -2-2-1 & 1+4+1 & -1-2-2 \\ 2+1+2 & -1-2-2 & 1+1+4 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$$

$$\therefore 4A^{-1} = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - 6 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 9 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6-12+9 & -5+6 & 5-6 \\ -5+6 & 6-12+9 & -5+6 \\ 5-6 & -5+6 & 6-12+9 \end{bmatrix} = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

$$(1) \Rightarrow A^3 = 6A^2 - 9A + 4I$$

$$\therefore A^4 = 6A^3 - 9A^2 + 4A \quad [\text{Multiplying by } A]$$

$$= 6[6A^2 - 9A + 4I] - 9A^2 + 4A$$

$$= 27A^2 - 50A + 24I$$

$$= 27 \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - 50 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} + 24 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 162 - 100 + 24 & -135 + 50 & 135 - 50 \\ -135 + 50 & 162 - 100 + 24 & -135 + 50 \\ 135 - 50 & -135 + 50 & 162 - 100 + 24 \end{bmatrix}$$

$$= \begin{bmatrix} 86 & -85 & 85 \\ -85 & 86 & -85 \\ 85 & -85 & 86 \end{bmatrix}$$

### Example 4

Use Cayley-Hamilton theorem to find the matrix

$$A^8 - 5A^7 + 7A^6 - 3A^5 + 8A^4 - 5A^3 + 8A^2 - 2A + I \text{ if the matrix } A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

[AU 2009, 2010]

### Solution

$$\text{Given } A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

The characteristic equation is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 1 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$\text{where } S_1 = 2 + 1 + 2 = 5$$

$$S_2 = \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix} = 2 + 4 - 1 + 2 = 7$$

$$S_3 = |A| = 2 \cdot 2 - 1 \cdot 0 + 1(-1) = 4 - 1 = 3$$

∴ the characteristic equation is  $\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$

By Cayley-Hamilton theorem, we get  $A^3 - 5A^2 + 7A - 3I = 0$

(1)

We have to find the matrix

$$A^8 - 5A^7 + 7A^6 - 3A^5 + 8A^4 - 5A^3 + 8A^2 - 2A + I = f(A), \text{ say}$$

We shall rewrite this matrix polynomial in terms of

$$A^3 - 5A^2 + 7A - 3I$$

∴ the polynomial

$$f(A) = A^5(A^3 - 5A^2 + 7A - 3I) + 8A^4 - 5A^3 + 8A^2 - 2A + I$$

$$= 8A^4 - 5A^3 + 8A^2 - 2A + I$$

[Using (1)]

$$= 8A(A^3 - 5A^2 + 7A - 3I) + 35A^3 - 48A^2 + 22A + I$$

$$= 35A^3 - 48A^2 + 22A + I$$

[Using (1)]

$$= 35(A^3 - 5A^2 + 7A - 3I) + 127A^2 - 223A + 106I$$

$$= 127A^2 - 223A + 106I$$

[Using (1)]

But  $A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix}$

$$\begin{aligned} \therefore f(A) &= 127 \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} - 223 \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + 106 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 635 - 446 + 106 & 508 - 223 & 508 - 223 \\ 0 & 127 - 223 + 106 & 0 \\ 508 - 223 & 508 - 223 & 635 - 446 + 106 \end{bmatrix} \\ &= \begin{bmatrix} 295 & 285 & 285 \\ 0 & 10 & 0 \\ 285 & 285 & 295 \end{bmatrix} \end{aligned}$$

### Example 5

Verify Cayley-Hamilton theorem and find the inverse of

$$A = \begin{bmatrix} 13 & -3 & 5 \\ 0 & 4 & 0 \\ -15 & 9 & -7 \end{bmatrix}$$

[AU 2006]

### Solution

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 13 - \lambda & -3 & 5 \\ 0 & 4 - \lambda & 0 \\ -15 & 9 & -7 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1 = 13 + 4 + (-7) = 10$

$$S_2 = \begin{vmatrix} 4 & 0 \\ 9 & -7 \end{vmatrix} + \begin{vmatrix} 13 & 5 \\ -15 & -7 \end{vmatrix} + \begin{vmatrix} 13 & -3 \\ 0 & 4 \end{vmatrix}$$

$$= -28 + (-91 + 75) + 52 - 0 = -28 - 16 + 52 = 8$$

$$S_3 = |A| = \begin{vmatrix} 13 & -3 & 5 \\ 0 & 4 & 0 \\ -15 & 9 & -7 \end{vmatrix}$$

$$= 4 \begin{vmatrix} 13 & 5 \\ -15 & -7 \end{vmatrix} = 4(-91 + 75) = -64 \quad [\text{expanding by } R_2]$$

$\therefore$  the characteristic equation is  $\lambda^3 - 10\lambda^2 + 8\lambda + 64 = 0$  (1)

By Cayley-Hamilton theorem A satisfies (1)

$\therefore$   $A^3 - 10A^2 + 8A + 64I = 0$  (2)

We shall now verify this by direct computations.

$$A^2 = A \cdot A = \begin{bmatrix} 13 & -3 & 5 \\ 0 & 4 & 0 \\ -15 & 9 & -7 \end{bmatrix} \begin{bmatrix} 13 & -3 & 5 \\ 0 & 4 & 0 \\ -15 & 9 & -7 \end{bmatrix}$$

$$= \begin{bmatrix} 94 & -6 & 30 \\ 0 & 16 & 0 \\ -90 & 18 & -26 \end{bmatrix}$$

$$A^3 = A^2 \cdot A = \begin{bmatrix} 94 & -6 & 30 \\ 0 & 16 & 0 \\ -90 & 18 & -26 \end{bmatrix} \begin{bmatrix} 13 & -3 & 5 \\ 0 & 4 & 0 \\ -15 & 9 & -7 \end{bmatrix}$$

$$= \begin{bmatrix} 772 & -36 & 260 \\ 0 & 64 & 0 \\ -780 & 108 & -268 \end{bmatrix}$$

Now  $A^3 - 10A^2 + 8A + 64I$

$$= \begin{bmatrix} 772 & -36 & 260 \\ 0 & 64 & 0 \\ -780 & 108 & -268 \end{bmatrix} - 10 \begin{bmatrix} 94 & -6 & 30 \\ 0 & 16 & 0 \\ -90 & 18 & -26 \end{bmatrix}$$

$$+ 8 \begin{bmatrix} 13 & -3 & 5 \\ 0 & 4 & 0 \\ -15 & 9 & -7 \end{bmatrix} + 64 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 772 - 940 + 104 + 64 & -36 + 60 - 24 + 0 & 260 - 300 + 40 + 0 \\ 0 + 0 + 0 + 0 & 64 - 160 + 32 + 64 & 0 + 0 + 0 + 0 \\ -780 + 900 - 120 + 0 & 108 - 180 + 72 + 0 & -268 + 260 - 56 + 64 \end{bmatrix}$$

$$= \begin{bmatrix} 940 - 940 & 60 - 60 & 300 - 300 \\ 0 & 160 - 160 & 0 \\ 900 - 900 & 180 - 180 & 324 - 324 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore A^3 - 10A^2 + 8A + 64I = 0$$

Hence the theorem is verified.

**To find  $A^{-1}$ :** We have  $64I = -A^3 + 10A^2 - 8A$

Multiplying by  $A^{-1}$ , we get

$$64A^{-1} = -A^2 + 10A - 8I \quad [\because AA^{-1} = I]$$

$$= - \begin{bmatrix} 94 & -6 & 30 \\ 0 & 16 & 0 \\ -90 & 18 & -26 \end{bmatrix} + 10 \begin{bmatrix} 13 & -3 & 5 \\ 0 & 4 & 0 \\ -15 & 9 & -7 \end{bmatrix} - 8 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -94 + 130 - 8 & 6 - 30 - 0 & -30 + 50 - 0 \\ 0 + 0 - 0 & -16 + 40 - 8 & 0 + 0 - 0 \\ 90 - 150 - 0 & -18 + 90 - 0 & 26 - 70 - 8 \end{bmatrix}$$

$$= \begin{bmatrix} 28 & -24 & 20 \\ 0 & 16 & 0 \\ -60 & 72 & -52 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{64} \begin{bmatrix} 28 & -24 & 20 \\ 0 & 16 & 0 \\ 60 & 72 & -52 \end{bmatrix}$$

$$= \frac{1}{16} \begin{bmatrix} 7 & -6 & 5 \\ 0 & 4 & 0 \\ 15 & 18 & -13 \end{bmatrix}$$

### Example 6

If  $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ , then show that  $A^n = A^{n-2} + A^2 - I$  for  $n \geq 3$ . Hence find

$A^{50}$ .

[AU 2006]

### Solution

Given  $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ .

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1 = 1, S_2 = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$

$$= -1 + 0 + 0 = -1$$

$$S_3 = |A| = -1$$

$\therefore$  the characteristic equation is  $\lambda^3 - \lambda^2 + \lambda + 1 = 0$

(1)

By Cayley-Hamilton theorem  $A$  satisfies (1)

$$\therefore A^3 - A^2 + A + I = 0$$

$$\Rightarrow A^3 - A^2 = A - I \quad (2)$$

Multiplying (2) by  $A, A^2, \dots, A^{n-3}$ , we get the equations

$$\begin{aligned} A^4 - A^3 &= A^2 - A \\ A^5 - A^4 &= A^3 - A^2 \\ A^6 - A^5 &= A^4 - A^3 \\ &\vdots \\ A^n - A^{n-1} &= A^{n-2} - A^{n-3} \end{aligned}$$

Adding (2) and all these equations we get

$$A^n - A^2 = A^{n-2} - I$$

$$\Rightarrow A^n = A^{n-2} + (A^2 - I) \text{ for all } n \geq 3 \quad (i)$$

$$\therefore A^{n-2} = A^{n-4} + A^2 - I$$

$$A^{n-4} = A^{n-6} + A^2 - I$$

$$\vdots$$

$$\begin{aligned} \therefore A^n &= (A^{n-4} + A^2 - I) + (A^2 - I) \\ &= A^{n-4} + 2(A^2 - I) \end{aligned} \quad (ii)$$

$$\begin{aligned} &= A^{n-6} + A^2 - I + 2(A^2 - I) \\ &= A^{n-6} + 3(A^2 - I) \end{aligned} \quad (iii)$$

$$\begin{aligned} &= A^{n-8} + 4(A^2 - I) \\ &\vdots \end{aligned} \quad (iv)$$

If  $n$  is even, then  $A^n = A^{n-(n-2)} + \left(\frac{n-2}{2}\right)(A^2 - I)$

[observe the coefficients of  $A^2 - I$  in (i), (ii), (iii) ... and index of  $A$ .

We see  $\frac{2}{2} = 1$  in (i),  $\frac{4}{2} = 2$  in (ii),  $\frac{6}{2} = 3$  in (iii),  $\frac{8}{2} = 4$  in (iv) and so on

$\frac{n-2}{2}$  in the last one]

$$\therefore A^n = A^2 + \left(\frac{n-2}{2}\right)A^2 - \frac{n-2}{2}I$$

$$A^n = \frac{n}{2}A^2 - \left(\frac{n-2}{2}\right)I$$

Putting  $n = 50$ , we get  $A^{50} = 25A^2 - 24I$

$$\begin{aligned} \text{But } A^2 &= A \cdot A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \end{aligned}$$

$$\therefore A^{50} = 25 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} - 24 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}$$

### Example 7

If  $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ , find  $A^n$  in terms of  $A$  and  $I$ .

### Solution

$$\text{Given } A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

The characteristic equation is  $\lambda^2 - 5 = 0$

[see example 17]

By Cayley-Hamiltons theorem  $A^2 - 5I = 0$

(1)

To find  $A^n$ , consider the polynomial  $\lambda^n$

Dividing  $\lambda^n$  by  $\lambda^2 - 5$  we get  $\lambda^n = (\lambda^2 - 5)\phi(\lambda) + a\lambda + b$

(2)

where  $\phi(\lambda)$  is the quotient and  $a\lambda + b$  is the remainder.

We shall now find the values of  $a$  and  $b$ .

The eigen values of  $A$  are  $\lambda = \sqrt{5}, -\sqrt{5}$

Substitute  $\lambda = \sqrt{5}$  in (2) then  $(\sqrt{5})^n = 0 + a\sqrt{5} + b$

$$\Rightarrow a\sqrt{5} + b = (\sqrt{5})^n \quad (3)$$

Substitute  $\lambda = -\sqrt{5}$  in (2) then  $(-\sqrt{5})^n = 0 + a(-\sqrt{5}) + b$

$$\Rightarrow -a\sqrt{5} + b = (-\sqrt{5})^n \quad (4)$$

$$(3) + (4) \Rightarrow 2b = (\sqrt{5})^n + (-\sqrt{5})^n$$

$$\therefore b = \frac{(\sqrt{5})^n + (-\sqrt{5})^n}{2} = (\sqrt{5})^n \frac{(1 + (-1)^n)}{2}$$

$$(3) - (4) \Rightarrow 2\sqrt{5}a = (\sqrt{5})^n - (-\sqrt{5})^n$$



$$\therefore a = \frac{(\sqrt{5})^n - (-\sqrt{5})^n}{2\sqrt{5}} = (\sqrt{5})^{n-1} \frac{1 - (-1)^n}{2}$$

Replacing  $\lambda$  by  $A$  in (2), we get

$$\begin{aligned} A^n &= (A^2 - 5I)\phi(A) + aA + bI \\ &= 0 + aA + bI \end{aligned}$$

[using (1)]

$$\therefore A^n = (\sqrt{5})^{n-1} \left( \frac{1 - (-1)^n}{2} \right) A + (\sqrt{5})^n \left( \frac{1 + (-1)^n}{2} \right) I.$$

### EXERCISE 1.2

Verify Cayley-Hamilton theorem for the following matrices and hence find their inverses.

1.  $\begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$

[AU 2000]

2.  $\begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

[AU 1998, 2005]

3.  $\begin{bmatrix} -1 & 0 & 3 \\ 8 & 1 & -7 \\ -3 & 0 & 8 \end{bmatrix}$

[AU 2004, 2010]

4.  $\begin{bmatrix} 7 & 2 & -2 \\ -6 & -1 & 2 \\ 6 & 2 & -1 \end{bmatrix}$

5. Verify that the matrix  $A = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$  satisfies its characteristic equation and hence find  $A^4$ . [AU 2008]

6.  $A = \begin{bmatrix} 7 & 3 \\ 2 & 6 \end{bmatrix}$ , find  $A^n$  in terms of  $A$  and  $I$  using Cayley-Hamilton theorem and hence find  $A^3$ .

7. Find  $A^4$  using Cayley-Hamilton theorem for the matrix  $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix}$ . Find  $A^4 + A^3 - 18A^2 - 39A + 2I$

[AU 2010, 2011]

8. Find the eigen values and eigen vectors of the system of equations  $10x_1 + 2x_2 + x_3 = \lambda x_1$ ,  $2x_1 + 10x_2 + x_3 = \lambda x_2$ ,  $2x_1 + x_2 + 10x_3 = \lambda x_3$   
[Hint: Equations can be rewritten as  $(10 - \lambda)x_1 + 2x_2 + x_3 = 0$ ,  $2x_1 + (10 - \lambda)x_2 + x_3 = 0$ ,  $2x_1 + x_2 + (10 - \lambda)x_3 = 0$  (I)]

If  $A = \begin{bmatrix} 10 & 2 & 1 \\ 2 & 10 & 1 \\ 2 & 1 & 10 \end{bmatrix}$  and  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  then these equations in matrix form

is  $(A - \lambda I)X = 0$  and so  $|A - \lambda I| = 0$  is the characteristic equation of  $A$  and  $\lambda = 8, 9, 13$ . Eigen vectors are given by (I)]

9. If  $\lambda$  is an eigen value of a non-singular matrix  $A$ , show that  $\frac{|A|}{\lambda}$  is an eigen value of the matrix  $\text{adj } A$ .

[Hint:  $AX = \lambda X \Rightarrow (\text{adj } A)(AX) = (\text{adj } A)(\lambda X)$

$$\Rightarrow |A|X = \lambda(\text{adj } A)X$$

$$\Rightarrow (\text{adj } A)X = \frac{|A|}{\lambda}X$$

10. Verify Cayley-Hamilton theorem for the matrix  $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$  and find its inverse and also find  $A^6 - 4A^5 + 8A^4 - 12A^3 + 14A^2$ .

### ANSWERS TO EXERCISE 1.2

$$1. \frac{1}{35} \begin{bmatrix} -4 & 11 & -5 \\ -1 & -6 & 25 \\ 6 & 1 & -10 \end{bmatrix} \quad 2. \frac{1}{9} \begin{bmatrix} 0 & 3 & 3 \\ 3 & 2 & -7 \\ 3 & -1 & -1 \end{bmatrix} \quad 3. \begin{bmatrix} 8 & 0 & -3 \\ -43 & 1 & 17 \\ 3 & 0 & -1 \end{bmatrix}$$

$$4. \frac{1}{3} \begin{bmatrix} -3 & -2 & 2 \\ 6 & 5 & -2 \\ -6 & -2 & 5 \end{bmatrix} \quad 5. A^4 = \begin{bmatrix} 124 & -123 & 162 \\ -95 & 96 & -123 \\ 95 & -95 & 124 \end{bmatrix}$$

$$6. A^n = \frac{9^n - 4^n}{5} \begin{bmatrix} 7 & 3 \\ 2 & 6 \end{bmatrix} + \frac{9 \cdot 4^n - 4 \cdot 9^n}{5} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; A^3 = \begin{bmatrix} 463 & 399 \\ 266 & 330 \end{bmatrix}$$

$$7. A^4 = \begin{bmatrix} 248 & 101 & 218 \\ 272 & 109 & 50 \\ 104 & 98 & 204 \end{bmatrix} \quad 8. \begin{bmatrix} 3 \\ -2 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$10. A^{-1} = \frac{1}{5} \begin{bmatrix} 3 & -2 \\ 1 & -1 \end{bmatrix}, \begin{bmatrix} 1 & -8 \\ 4 & -7 \end{bmatrix}$$

## 1.4 SIMILARITY TRANSFORMATION AND ORTHOGONAL TRANSFORMATION

### 1.4.1 Similar matrices

**Definition 1.4.1** Let  $A$  and  $B$  be square matrices of order  $n$ .  $A$  is said to be similar to  $B$  if there exists a non-singular matrix  $P$  of order  $n$  such that

$$A = P^{-1}BP \quad (1)$$

The transformation (1) which transforms  $B$  into  $A$  is called a similarity transformation. The matrix  $P$  is called a similarity matrix.

**Note** We shall now see that if  $A$  is similar to  $B$  then  $B$  is similar to  $A$ .

$$A = P^{-1}BP \Rightarrow PA P^{-1} = B$$

(Premultiplying by  $P$  and postmultiplying by  $P^{-1}$ )

$$\Rightarrow (P^{-1})^{-1} A (P^{-1}) = B$$

$$\Rightarrow Q^{-1}AQ = B \quad (2)$$

where  $Q = P^{-1}$

The relation (2) means  $B$  is similar to  $A$ . Thus, if  $A$  is similar to  $B$ , then  $B$  is similar to  $A$ . Hence we simply say similar matrices.

An important property of similarity transformations is that they preserve eigen values, which is proved in the next theorem.

### Theorem 1.4.1 Similar matrices have the same eigen values.

**Proof** Let  $A$  and  $B$  be two similar matrices of order  $n$ .

Then  $B = P^{-1}AP$ , by definition.

$\therefore$  the characteristic polynomial of  $B$  is  $|B - \lambda I|$

$$\begin{aligned} \text{Now } |B - \lambda I| &= |P^{-1}AP - \lambda I| \\ &= |P^{-1}AP - \lambda P^{-1}IP| \\ &= |P^{-1}(A - \lambda I)P| \\ &= |P^{-1}| |A - \lambda I| |P| \quad [\because |AB| = |A| |B|] \\ &= |A - \lambda I| |P^{-1}| |P| = |A - \lambda I| |P^{-1}P| \\ &= |A - \lambda I| |I| \\ \Rightarrow |B - \lambda I| &= |A - \lambda I| \quad [\because |I| = 1] \end{aligned}$$

$\therefore$   $A$  and  $B$  have the same characteristic polynomial and hence have the same characteristic equation. So  $A$  and  $B$  have the same eigen values. ■

**Note** Similar matrices  $A$  and  $B$  have the same determinant value i.e.,  $|A| = |B|$ .

$$\begin{aligned} \text{For } B = P^{-1}AP \Rightarrow |B| &= |P^{-1}AP| \\ &= |P^{-1}| |A| |P| \\ &= |A| |P^{-1}P| \\ &= |A| |I| = |A| \end{aligned}$$

### 1.4.2 Diagonalisation of a square matrix

**Definition 1.4.2** A square matrix  $A$  is said to be **diagonalisable** if there exists a non-singular matrix  $P$  such that  $P^{-1}AP = D$ , where  $D$  is a diagonal matrix. The matrix  $P$  is called a model matrix of  $A$ .

The next theorem provides us with a method of diagonalisation.

**Theorem 1.4.2** If  $A$  is a square matrix of order  $n$ , having  $n$  linearly independent eigen vectors and  $M$  is the matrix whose columns are the eigen vectors of  $A$ , then  $M^{-1}AM = D$ , where  $D$  is the diagonal matrix whose diagonal elements are the eigen values of  $A$ .

**Proof** Let  $X_1, X_2, \dots, X_n$  be  $n$  linearly independent eigen vectors of  $A$  corresponding to the eigen values  $\lambda_1, \lambda_2, \dots, \lambda_n$  of  $A$ .

$$\therefore AX_i = \lambda_i X_i, \quad i = 1, 2, 3, \dots, n.$$

Let  $M = [X_1 \ X_2 \ \dots \ X_n]$  be the matrix formed with the eigen vectors as columns.

Then  $AM = [AX_1 \ AX_2 \ AX_3 \ \dots \ AX_n]$

$$= [\lambda_1 X_1 \ \lambda_2 X_2 \ \lambda_3 X_3 \ \dots \ \lambda_n X_n]$$

$$= [X_1 \ X_2 \ \dots \ X_n] \begin{bmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & \dots \\ 0 & 0 & \lambda_3 & \dots \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \lambda_n \end{bmatrix}$$

$$AM = MD, \quad \text{where } D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ \vdots & \vdots & \vdots \\ 0 & \dots & \lambda_n \end{bmatrix}$$

$$M^{-1}AM = D$$

The matrix  $M$  which diagonalises  $A$  is called the **model matrix** of  $A$  and the resulting diagonal matrix  $D$  whose elements are eigen values of  $A$  is called the **spectral matrix** of  $A$ . ■

### 1.4.3 Computation of powers of a square matrix

Diagonalisation of a square matrix  $A$  is very useful to find powers of  $A$ , say  $A^r$ .

By the theorem 1.4.2,  $D = M^{-1}AM$

$$\begin{aligned} \therefore D^2 &= (M^{-1}AM)(M^{-1}AM) \\ &= M^{-1}A(MM^{-1})AM \\ &= M^{-1}AIAM = M^{-1}A^2M \end{aligned}$$

Similarly,

$$\begin{aligned}
 D^3 &= D^2 D \\
 &= (M^{-1} A^2 M) (M^{-1} A M) \\
 &= M^{-1} A^2 (M M^{-1}) A M \\
 D^3 &= M^{-1} A^2 I A M = M^{-1} A^3 M
 \end{aligned}$$

Proceeding in this way, we can find

$$D^r = M^{-1} A^r M$$

$$\therefore A^r = M D^r M^{-1}, \text{ where } D^r = \begin{bmatrix} \lambda_1^r & 0 & \dots & 0 \\ 0 & \lambda_2^r & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_n^r \end{bmatrix}$$

### Note

- (1) If the eigen values  $\lambda_1, \lambda_2, \dots, \lambda_n$  of  $A$  are different then the corresponding eigen vectors  $X_1, X_2, \dots, X_n$  are linearly independent by theorem 1.2.1 (2). So  $A$  can be diagonalised.
- (2) Even if 2 or more eigen values are equal, if we can find independent eigen vectors corresponding to them (see worked example 6), then  $A$  can be diagonalised.

Thus independence of eigen vectors is the condition for diagonalisation.

### Working rule to diagonalise a $n \times n$ matrix $A$ by similarity transformation:

**Step 1:** Find the eigen values  $\lambda_1, \lambda_2, \dots, \lambda_n$

**Step 2:** Find linearly independent eigen vectors  $X_1, X_2, \dots, X_n$

**Step 3:** Form the modal matrix  $M = [X_1 \ X_2 \ \dots \ X_n]$

**Step 4:** Find  $M^{-1}$  and  $AM$ .

$$\text{Step 5: Compute } M^{-1} A M = D = \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & \dots & \lambda_n \end{bmatrix}$$

### 1.4.4 Orthogonal matrix

**Definition 1.4.3** A real square matrix  $A$  is said to be orthogonal if  $AA^T = A^T A = I$ , where  $I$  is the unit matrix of the same order as  $A$ .

From this definition it is clear that  $A^T = A^{-1}$ . So an orthogonal matrix is also defined as below.

**Definition 1.4.4** A real square matrix  $A$  is orthogonal if  $A^T = A^{-1}$ .

**Example 1**

Prove that  $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$  is orthogonal. [AU 2001, 2009]

**Solution**

Given  $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$

$$\therefore |A| = \begin{vmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{vmatrix} = \cos^2 \theta + \sin^2 \theta = 1$$

$\therefore A$  is non-singular.

$$\begin{aligned} \text{Hence } A^{-1} &= \frac{\text{adj } A}{|A|} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}^T \\ &= A^T \end{aligned} \quad [\because |A| = 1]$$

$\therefore A$  is orthogonal.

**Properties of orthogonal matrix**

**1. If  $A$  is orthogonal, then  $A^T$  is orthogonal.**

**Proof** Given  $A$  is orthogonal.  $\therefore AA^T = A^T A = I$   
Reversing the roles of  $A$  and  $A^T$ , we see  $A^T A = AA^T = I$   
 $\Rightarrow A^T$  is orthogonal. ■

**Note** Since  $A^T = A^{-1}$ , it follows  $A^{-1}$  is orthogonal.

**2. If  $A$  is an orthogonal matrix, then  $|A| = \pm 1$ .**

**Proof** Given  $A$  is orthogonal. Then  $AA^T = I$

$$\Rightarrow |AA^T| = 1$$

$$\Rightarrow |A| |A^T| = 1$$

$$\text{But we know } |A^T| = |A|$$

$$\therefore |A| |A| = 1$$

$$\Rightarrow |A|^2 = 1 \Rightarrow |A| = \pm 1 \quad \blacksquare$$

**3. If  $\lambda$  is an eigen value of an orthogonal matrix  $A$ , then  $\frac{1}{\lambda}$  is also an eigen value of  $A$ .**

**Proof** Given  $A$  is orthogonal and  $\lambda$  is an eigen value of  $A$ .

Then  $A^T = A^{-1}$ . By property (4) of eigen values,

$\frac{1}{\lambda}$  is an eigen value of  $A^{-1}$  and so an eigen value of  $A^T$ .

By property (1),  $A$  and  $A^T$  have same eigen values.

$\therefore \frac{1}{\lambda}$  is an eigen of  $A$  and hence  $\lambda, \frac{1}{\lambda}$  are eigen values of orthogonal matrix  $A$ . ■

#### 4. If $A$ and $B$ are orthogonal matrices, then $AB$ is orthogonal.

**Proof** Given  $A$  and  $B$  are orthogonal matrices.

$$\therefore A^T = A^{-1} \text{ and } B^T = B^{-1}$$

Now

$$\begin{aligned} (AB)^T &= B^T A^T \\ &= B^{-1} A^{-1} \\ &= (AB)^{-1} \end{aligned}$$

$$\therefore AB \text{ is orthogonal.} \quad \blacksquare$$

#### 5. Eigen values of an orthogonal matrix are of magnitude 1.

**Proof** Let  $A$  be an orthogonal matrix and let  $\lambda$  be an eigen value of  $A$ .

Then  $AX = \lambda X$ , where  $X \neq 0$

(1)

Taking complex conjugate, we get  $\overline{A} \overline{X} = \overline{\lambda} \overline{X}$

But  $A$  is real matrix  $\therefore \overline{A} = A$

Hence  $A \overline{X} = \overline{\lambda} \overline{X}$

Taking transpose,  $(A \overline{X})^T = (\overline{\lambda} \overline{X})^T$

$$\Rightarrow \overline{X}^T A^T = \overline{\lambda} \overline{X}^T$$

$$\Rightarrow \overline{X}^T A^{-1} = \overline{\lambda} \overline{X}^T \quad [\because A^T = A^{-1}] \quad (2)$$

Multiplying (2) and (1) we get

$$(\overline{X}^T A^{-1})(AX) = (\overline{\lambda} \overline{X}^T)(\lambda X)$$

$$\Rightarrow \overline{X}^T (A^{-1}A)X = \overline{\lambda} \lambda \overline{X}^T X$$

$$\Rightarrow \overline{X}^T X = |\lambda|^2 \overline{X}^T X \quad [\because \overline{X}^T X \neq 0 \text{ as } X \neq 0]$$

$$\Rightarrow 1 = |\lambda|^2$$

$$\therefore |\lambda| = 1$$

This is true for all eigen values of  $A$ .

Hence eigen values of  $A$  are of absolute value 1. ■

### 1.4.5 Symmetric matrix

**Definition 1.4.5** A real square matrix  $A$  is said to be symmetric if  $A^T = A$ .

e.g.,  $\begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix}$ ,  $\begin{bmatrix} 1 & -1 & 3 \\ -1 & 0 & 4 \\ 3 & 4 & 2 \end{bmatrix}$  are symmetric matrices.

Note that the elements equidistant from the main diagonal are same.

### Properties of symmetric matrices

#### 1. Eigen values of a symmetric matrix are real.

**Proof** Let  $A$  be a symmetric matrix of order  $n$  and  $\lambda$  be an eigen value of  $A$ .

Then there exists  $X \neq 0$  such that  $AX = \lambda X$

(1)

Taking complex conjugate,

$$\overline{A} \overline{X} = \overline{\lambda} \overline{X}$$

$$\Rightarrow A \overline{X} = \overline{\lambda} \overline{X} \quad [\because A \text{ is real } \overline{A} = A] \quad (2)$$

$$\text{Taking transpose, } (A \overline{X})^T = (\overline{\lambda} \overline{X})^T$$

$$\Rightarrow \overline{X}^T A^T = \overline{\lambda} \overline{X}^T$$

$$\Rightarrow \overline{X}^T A = \overline{\lambda} \overline{X}^T \quad [\because A^T = A]$$

Post multiplying by  $X$ ,

$$(\overline{X}^T A)X = (\overline{\lambda} \overline{X}^T)X$$

$$\Rightarrow \overline{X}^T (AX) = \overline{\lambda} (\overline{X}^T X)$$

$$\Rightarrow \overline{X}^T (\lambda X) = \overline{\lambda} (\overline{X}^T X)$$

$$\Rightarrow \lambda (\overline{X}^T X) = \overline{\lambda} (\overline{X}^T X)$$

$$\Rightarrow \lambda = \overline{\lambda} \quad [\because \overline{X}^T X \neq 0 \text{ as } X \neq 0]$$

$\therefore \lambda$  is real. This is true for all eigen values.

$\therefore$  eigen values of a symmetric matrix are real. ■

#### 2. Eigen vectors corresponding to different eigen values of a symmetric matrix are orthogonal vectors.

**Proof** Let  $A$  be a symmetric matrix of order  $n$ .

$$\therefore A^T = A.$$



Let  $\lambda_1, \lambda_2$  be two different eigen values of A.

Then  $\lambda_1, \lambda_2$  are real, by property (1).

$\therefore$  there exist  $X_1 \neq 0, X_2 \neq 0$  such that

$$AX_1 = \lambda_1 X_1 \quad (1)$$

and

$$AX_2 = \lambda_2 X_2 \quad (2)$$

Premultiplying (1) by  $X_2^T$ , we get  $X_2^T(AX_1) = X_2^T \lambda_1 X_1$

$$\Rightarrow X_2^T AX_1 = \lambda_1 (X_2^T X_1) \quad (3)$$

Premultiplying (2) by  $X_1^T$ , we get  $X_1^T(AX_2) = X_1^T \lambda_2 X_2$

$$\Rightarrow X_1^T AX_2 = \lambda_2 (X_1^T X_2) \quad (4)$$

Taking transpose of (3), we get  $(X_2^T AX_1)^T = (\lambda_1 X_2^T X_1)^T$

$$\Rightarrow X_1^T A^T X_2 = \lambda_1 X_1^T X_2$$

$$\Rightarrow X_1^T AX_2 = \lambda_1 (X_1^T X_2) \quad (5)$$

From (4) and (5) we get,  $\lambda_1 (X_1^T X_2) = \lambda_2 (X_1^T X_2)$

$$\Rightarrow (\lambda_1 - \lambda_2)(X_1^T X_2) = 0$$

Since  $\lambda_1 \neq \lambda_2, \lambda_1 - \lambda_2 \neq 0, \therefore X_1^T X_2 = 0$

$\Rightarrow X_1$  and  $X_2$  are orthogonal. ■

**Remark** If  $X_1 = (a_1, b_1, c_1)$  and  $X_2 = (a_2, b_2, c_2)$  be two 3-dimensional vectors, they are orthogonal if their dot product is 0  $\Rightarrow a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

If we treat them as column matrices,  $X_1 = \begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix}$   $X_2 = \begin{bmatrix} a_2 \\ b_2 \\ c_2 \end{bmatrix}$ , then the matrix

$$\begin{aligned} \text{product } X_1^T X_2 &= [a_1 \ b_1 \ c_1] \begin{bmatrix} a_2 \\ b_2 \\ c_2 \end{bmatrix} \\ &= a_1 a_2 + b_1 b_2 + c_1 c_2 \end{aligned}$$

So  $X_1$  and  $X_2$  are orthogonal if  $X_1^T X_2 = 0$  or  $X_2^T X_1 = 0$ .

Thus we can treat column matrices as vectors and verify dot product = 0.

2. The unit vector in  $X_1$  is  $\frac{X_1}{\sqrt{a^2 + b^2 + c^2}}$  and it is called a **normalised vector**.

**Note** For any square matrix eigen vectors corresponding to different eigen values are linearly independent, but for a symmetric matrix, they are orthogonal, pairwise.

### 1.4.6 Diagonalisation by orthogonal transformation or orthogonal reduction

**Definition 1.4.6** A square matrix  $A$  is said to be orthogonally diagonalisable if there exists an orthogonal matrix  $N$  such that

$$N^{-1}AN = D \Rightarrow N^TAN = D \quad [\because N^T = N^{-1}]$$

This transformation which transforms  $A$  into  $D$  is called an orthogonal transformation or orthogonal reduction.

The next theorem gives a method of orthogonal reduction.

**Theorem 1.4.3** Let  $A$  be a symmetric matrix of order  $n$ . Let  $X_1, X_2, \dots, X_n$  be eigen vectors of  $A$  which are pairwise orthogonal. Let  $N$  be the matrix formed with the normalised eigen vectors of  $A$  as columns. Then  $N$  is an orthogonal matrix such that  $N^{-1}AN = D \Rightarrow N^TAN = D$ .

$N$  is called normalised modal matrix of  $A$  or normal modal matrix of  $A$ .

#### Working rule for orthogonal reduction of a $n \times n$ symmetric matrix.

**Step 1:** Find the eigen values  $\lambda_1, \lambda_2, \dots, \lambda_n$

**Step 2:** Find the eigen vectors  $X_1, X_2, \dots, X_n$  which are pairwise orthogonal.

**Step 3:** Form the normalised modal matrix  $N$  with the normalised eigen vectors as columns.

**Step 4:** Find  $N^T$  and  $AN$ .

**Step 5:** Compute  $N^TAN = D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ \vdots & \dots & \dots \\ 0 & \dots & \lambda_n \end{bmatrix}$

### WORKED EXAMPLES

#### Example 1

**Diagonalise the matrix  $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix}$  by means of an orthogonal transformation.** [AU 2004, 2011]

**Solution**

A is a symmetric matrix. The characteristic equation of A is  $|A - \lambda I| = 0$   
 $\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$

where  $S_1 = 3 + 3 + 3 = 9$

$$S_2 = \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} + \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} + \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix}$$

$$= (9 - 1) + (9 - 1) + (9 - 1) = 24$$

$$S_3 = \begin{vmatrix} 3 & 1 & 1 \\ 1 & 3 & -1 \\ 1 & -1 & 3 \end{vmatrix}$$

$$= 3(9 - 1) - 1(3 + 1) + (-1 - 3) = 24 - 4 - 4 = 16$$

$\therefore$  The characteristic equation is  $\lambda^3 - 9\lambda^2 + 24\lambda - 16 = 0$

By trial we find  $\lambda = 1$  is a root.

Other roots are given by  $\lambda^2 - 8\lambda + 16 = 0$

$$\Rightarrow (\lambda - 4)^2 = 0$$

$$\Rightarrow \lambda = 4, 4$$

$\therefore$  the eigen values are  $\lambda = 1, 4, 4$ .

$$1 \quad \begin{array}{ccc|c} 1 & -9 & 24 & -16 \\ 0 & 1 & -8 & 16 \\ 1 & -8 & 16 & 0 \end{array}$$

**To find eigen vectors:**

Let  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector corresponding  $\lambda$ .

$$\text{Then } (A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} 3-\lambda & 1 & 1 \\ 1 & 3-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left. \begin{array}{l} (3-\lambda)x_1 + x_2 + x_3 = 0 \\ x_1 + (3-\lambda)x_2 - x_3 = 0 \\ x_1 - x_2 + (3-\lambda)x_3 = 0 \end{array} \right\} \quad (I)$$

**Case (i)** If  $\lambda = 1$  then equations (I) become

$$2x_1 + x_2 + x_3 = 0$$

$$x_1 + 2x_2 - x_3 = 0$$

$$x_1 - x_2 + 2x_3 = 0$$

Choosing the first two equations, we have

$$\begin{aligned} \frac{x_1}{-1-2} &= \frac{x_2}{1+2} = \frac{x_3}{4-1} \\ \Rightarrow \frac{x_1}{-3} &= \frac{x_2}{3} = \frac{x_3}{3} \\ \Rightarrow \frac{x_1}{-1} &= \frac{x_2}{1} = \frac{x_3}{1} \end{aligned}$$

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ \begin{array}{c} 1 \\ 2 \end{array} \begin{array}{c} \nearrow \\ \searrow \end{array} & \begin{array}{c} 1 \\ -1 \end{array} \begin{array}{c} \nearrow \\ \searrow \end{array} & \begin{array}{c} 2 \\ 1 \end{array} \begin{array}{c} \nearrow \\ \searrow \end{array} \end{array} \begin{array}{c} 1 \\ 2 \end{array}$$

Choosing  $x_1 = -1, x_2 = 1, x_3 = 1$ , we get an eigen vector

$$X_1 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

**Case (ii)** If  $\lambda = 4$ , then the equations (I) become

$$\begin{aligned} -x_1 + x_2 + x_3 &= 0 \Rightarrow x_1 - x_2 - x_3 = 0 \\ x_1 - x_2 - x_3 &= 0 \\ x_1 - x_2 - x_3 &= 0 \end{aligned}$$

We get only one equation  $x_1 - x_2 - x_3 = 0$  (1)

To solve for  $x_1, x_2, x_3$ , we can assign arbitrary values for two of the variables and we shall find 2 orthogonal vectors.

Put  $x_3 = 0, x_2 = 1$ , then  $x_1 = 1$ , we get an eigen vector  $X_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ .

Let  $X_3 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$  be  $\perp$  to  $X_2$ . Then  $a + b = 0 \Rightarrow b = -a$  and  $X_3$  should

satisfy (1)  $\therefore a - b - c = 0$

Choosing  $a = 1$ , we get  $b = -1$  and  $c = 2$

$$\therefore X_3 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

Thus the eigen values are  $\lambda = 1, 4, 4$ , and the corresponding eigen vector are

$$X_1 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad X_3 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

The normalised eigen vectors are

$$\begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \quad \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \quad \begin{bmatrix} \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}$$

So, the normalised modal matrix  $N = \begin{bmatrix} \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{bmatrix}$

$$\therefore N^T = \begin{bmatrix} \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{bmatrix}$$

$$AN = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} \frac{-1}{\sqrt{3}} & \frac{4}{\sqrt{2}} & \frac{4}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{4}{\sqrt{2}} & \frac{-4}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{8}{\sqrt{6}} \end{bmatrix}$$

$$\begin{aligned} \therefore N^T AN &= \begin{bmatrix} \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} \frac{-1}{\sqrt{3}} & \frac{4}{\sqrt{2}} & \frac{4}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{4}{\sqrt{2}} & \frac{-4}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{8}{\sqrt{6}} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}, \text{ which is a diagonal matrix.} \end{aligned}$$

### Example 2

Diagonalise the symmetric matrix  $\begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & -2 \\ -1 & -2 & 1 \end{bmatrix}$  by an orthogonal transformation.

### Solution

Let  $A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & -2 \\ -1 & -2 & 1 \end{bmatrix}$ .

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The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 1 & -1 \\ 1 & 1-\lambda & -2 \\ -1 & -2 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$\text{where } S_1 = 2 + 1 + 1 = 4$$

$$S_2 = \begin{vmatrix} 1 & -2 \\ -2 & 1 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ -1 & 1 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix}$$

$$= -3 + 1 + 1 = -1$$

$$S_3 = |A| = 2(1-4) - 1(1-2) - (-2+1)$$

$$= -6 + 1 + 1 = -4$$

$\therefore$  the characteristic equation is  $\lambda^3 - 4\lambda^2 - \lambda + 4 = 0$

By trial  $\lambda = 1$  is a root.

Other roots are given by  $\lambda^2 - 3\lambda - 4 = 0$

$$\Rightarrow (\lambda - 4)(\lambda + 1) = 0$$

$$\Rightarrow \lambda = 4, -1$$

$$\begin{array}{r|rrrr} 1 & 1 & -4 & -1 & 4 \\ & 0 & 1 & -3 & -4 \\ \hline & 1 & -3 & -4 & 0 \end{array}$$

$\therefore$  the eigen values of A are  $\lambda = -1, 1, 4$

**To find eigen vectors:**

Let  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector corresponding to the eigen value  $\lambda$ .

$$\text{Then } (A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} 2-\lambda & 1 & -1 \\ 1 & 1-\lambda & -2 \\ -1 & -2 & 1-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left. \begin{aligned} (2-\lambda)x_1 + x_2 - x_3 &= 0 \\ x_1 + (1-\lambda)x_2 - 2x_3 &= 0 \\ -x_1 - 2x_2 + (1-\lambda)x_3 &= 0 \end{aligned} \right\} \quad (I)$$

**Case (i)** If  $\lambda = -1$ , then the equations (I) become

$$3x_1 + x_2 - x_3 = 0$$

$$x_1 + 2x_2 - 2x_3 = 0$$

$$-x_1 - 2x_2 + 2x_3 = 0 \Rightarrow x_1 + 2x_2 - 2x_3 = 0$$

From the first two equations we get

$$\begin{aligned} \frac{x_1}{-2+2} &= \frac{x_2}{-1+6} = \frac{x_3}{6-1} \\ \Rightarrow \frac{x_1}{0} &= \frac{x_2}{5} = \frac{x_3}{5} \\ \Rightarrow \frac{x_1}{0} &= \frac{x_2}{1} = \frac{x_3}{1} \end{aligned}$$

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ \begin{array}{cc} 1 & -1 \\ 2 & -2 \end{array} & \begin{array}{cc} 3 & 1 \\ -2 & 1 \end{array} & \begin{array}{cc} 1 & 1 \\ 2 & 2 \end{array} \end{array}$$

Choosing  $x_1 = 0, x_2 = 1, x_3 = 1$ , we get an eigen vector,

$$X_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

**Case (ii)** If  $\lambda = 1$ , the equations (I) become

$$\begin{aligned} x_1 + x_2 - x_3 &= 0 \\ x_1 + 0x_2 - 2x_3 &= 0 \\ -x_1 - 2x_2 + 0x_3 &= 0 \end{aligned}$$

From the first two equations, we get

$$\begin{aligned} \frac{x_1}{-2-0} &= \frac{x_2}{-1+2} = \frac{x_3}{0-1} \\ \Rightarrow \frac{x_1}{-2} &= \frac{x_2}{1} = \frac{x_3}{-1} \\ \Rightarrow \frac{x_1}{2} &= \frac{x_2}{-1} = \frac{x_3}{1} \end{aligned}$$

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ \begin{array}{cc} 1 & -1 \\ 0 & -2 \end{array} & \begin{array}{cc} 1 & 1 \\ -2 & 1 \end{array} & \begin{array}{cc} 1 & 1 \\ 0 & 0 \end{array} \end{array}$$

Choosing  $x_1 = 2, x_2 = -1, x_3 = 1$ , we get an eigen vector,

$$X_2 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

**Case (iii)** If  $\lambda = 4$ , the equations (I) become

$$\begin{aligned} -2x_1 + x_2 - x_3 &= 0 \\ x_1 - 3x_2 - 2x_3 &= 0 \\ -x_1 - 2x_2 - 3x_3 &= 0 \end{aligned}$$

From the first and second equations, we get

$$\begin{aligned} \frac{x_1}{-2-3} &= \frac{x_2}{-1-4} = \frac{x_3}{6-1} \\ \Rightarrow \frac{x_1}{-5} &= \frac{x_2}{-5} = \frac{x_3}{5} \\ \Rightarrow \frac{x_1}{1} &= \frac{x_2}{1} = \frac{x_3}{-1} \end{aligned}$$

$$\begin{array}{ccccc} & x_1 & & x_2 & & x_3 \\ 1 & \swarrow \searrow & -1 & \swarrow \searrow & -2 & \swarrow \searrow & 1 \\ -3 & & -2 & & 1 & & -3 \end{array}$$

Choosing vector  $x_1 = 1, x_2 = 1, x_3 = -1$ , we get an eigen vector

$$X_3 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Thus the eigen values of A are  $-1, 1, 4$  and the corresponding eigen vectors are

$$X_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad X_3 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Since A is symmetric, the eigen vectors are pairwise orthogonal.  
The normalised eigen vectors are

$$\begin{bmatrix} 0 \\ 1 \\ 1 \\ \sqrt{2} \end{bmatrix}, \quad \begin{bmatrix} 2 \\ \sqrt{6} \\ -1 \\ \sqrt{6} \end{bmatrix}, \quad \begin{bmatrix} 1 \\ \sqrt{3} \\ 1 \\ \sqrt{3} \end{bmatrix}$$

So, the normalised modal matrix N =

$$\begin{bmatrix} 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{3}} \end{bmatrix}$$

$\therefore$

$$N^T = \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{3}} \end{bmatrix}$$



$$\begin{aligned}
 AN &= \begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & -2 \\ -1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{3}} \end{bmatrix} \\
 &= \begin{bmatrix} 0 & \frac{2}{\sqrt{6}} & \frac{4}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{4}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{-4}{\sqrt{3}} \end{bmatrix} \\
 \therefore N^T AN &= \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} 0 & \frac{2}{\sqrt{6}} & \frac{4}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{4}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{4}{\sqrt{3}} \end{bmatrix} \\
 &= \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix} = D
 \end{aligned}$$

### Example 3

The eigen vectors of a  $3 \times 3$  real symmetric matrix **A** corresponding to the eigen values 2, 3, 6 are  $[1, 0, -1]^T$ ,  $[1, 1, 1]^T$ ,  $[-1, 2, -1]^T$  respectively, find the matrix **A**. [AU 2002, 2006]

### Solution

Given **A** is symmetric and the eigen values are different.

So, the eigen vectors are orthogonal pairwise.

The normalised eigen vectors are

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \end{bmatrix}$$

So, the normalised modal matrix is  $N = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \end{bmatrix}$

Then by orthogonal reduction theorem 1.4.3

$$N^T A N = D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}, \text{ since 2, 3, 6 are the eigen values.}$$

But  $N^T = N^{-1}$

$$\therefore N^{-1} A N = D \Rightarrow A = N D N^{-1} = N D N^T$$

$$\begin{aligned} \Rightarrow A &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{bmatrix} \\ &= \begin{bmatrix} \sqrt{2} & \sqrt{3} & -\sqrt{6} \\ 0 & \sqrt{3} & 2\sqrt{6} \\ -\sqrt{2} & \sqrt{3} & -\sqrt{6} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{bmatrix} \\ &= \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix} \text{ which is the required matrix.} \end{aligned}$$

#### Example 4

**Diagonalise the matrix  $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ . Hence find  $A^3$ .**

[AU 2006]

#### Solution

Given  $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$  which is a symmetric matrix.

So, we shall diagonalise by orthogonal transformation.

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 0 & 1 \\ 0 & 3-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1 = 2 + 3 + 2 = 7$

$$S_2 = \begin{vmatrix} 3 & 0 \\ 0 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} = 6 + 3 + 6 = 15$$

$$S_3 = |A| = 2(6) - 0 + 1(-3) = 12 - 3 = 9$$

$\therefore$  the characteristic equation is  $\lambda^3 - 7\lambda^2 + 15\lambda - 9 = 0$

By trial  $\lambda = 1$  is a root.

Other roots are given by  $\lambda^2 - 6\lambda + 9 = 0$

$$\Rightarrow (\lambda - 3)^2 = 0$$

$$\Rightarrow \lambda = 3, 3$$

$\therefore$  the eigen values are  $\lambda = 1, 3, 3$

$$1 \quad \begin{vmatrix} 1 & -7 & 15 & -9 \\ 0 & 1 & -6 & 9 \\ 1 & -6 & 9 & 0 \end{vmatrix}$$

**To find eigen vectors:**

Let  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector corresponding to eigen value  $\lambda$ .

$$\text{Then } (A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} 2-\lambda & 0 & 1 \\ 0 & 3-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left. \begin{aligned} (2-\lambda)x_1 + 0x_2 + x_3 &= 0 \\ 0x_1 + (3-\lambda)x_2 + 0x_3 &= 0 \\ x_1 + 0x_2 + (2-\lambda)x_3 &= 0 \end{aligned} \right\} \quad (I)$$

**Case (i)** If  $\lambda = 1$ , then equations (I) become

$$x_1 + x_3 = 0 \Rightarrow x_3 = -x_1$$

$$2x_2 = 0 \Rightarrow x_2 = 0$$

$$x_1 + x_3 = 0$$

Choose  $x_1 = 1 \quad \therefore x_3 = -1$

$$\therefore \text{ an eigen vector is } X_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

**Case (ii)** If  $\lambda = 3$ , then equations (I) become

$$\left. \begin{aligned} -x_1 + x_3 &= 0 \\ 0x_2 &= 0 \\ x_1 - x_3 &= 0 \Rightarrow x_1 = x_3 \end{aligned} \right\} \quad (2)$$

and  $x_2$  can take any value  
Choosing  $x_1 = 1$ , we get  $x_3 = 1$  and choose  $x_2 = 0$

$\therefore$  an eigen vector is  $X_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

We shall now choose  $X_3 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$  orthogonal to  $X_2$

$\therefore$  dot product  $= 0 \Rightarrow a + c = 0$  and  $X_3$  should satisfy equations (2)

$\therefore a - c = 0$  and  $0b = 0$

Solving, we get  $a = c = 0$  and choose  $b = 1$

$\therefore X_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

Thus the eigen values are 1, 3, 3 and the corresponding eigen vectors are

$$\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Clearly they are pairwise orthogonal vectors.

The normalised eigen vectors are  $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \quad \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$\therefore$  the normalised modal matrix  $N = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$

$$\therefore N^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \end{bmatrix}$$

$$AN = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{3}{\sqrt{2}} & 0 \\ 0 & 0 & 3 \\ -\frac{1}{\sqrt{2}} & \frac{3}{\sqrt{2}} & 0 \end{bmatrix}$$

$$\begin{aligned} \therefore N^T AN &= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{3}{\sqrt{2}} & 0 \\ 0 & 0 & 3 \\ -\frac{1}{\sqrt{2}} & \frac{3}{\sqrt{2}} & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} = D \end{aligned}$$

$$\therefore N^{-1}AN = D \Rightarrow A = ND N^{-1}$$

$$\therefore A^3 = ND^3 N^{-1} = ND^3 N^T$$

$$\begin{aligned} \Rightarrow A^3 &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{-1}{\sqrt{2}} \\ 27 & 0 & \frac{27}{\sqrt{2}} \\ \frac{\sqrt{2}}{0} & 27 & \frac{\sqrt{2}}{0} \end{bmatrix} = \begin{bmatrix} 14 & 0 & 13 \\ 0 & 27 & 0 \\ 13 & 0 & 14 \end{bmatrix} \end{aligned}$$

## 1.5 REAL QUADRATIC FORM. REDUCTION TO CANONICAL FORM

**Definition 1.5.1** A homogeneous polynomial of second degree in any number of variables is called a quadratic form.

For example

- (i)  $x^2 + 4xy + 4y^2$   
 (ii)  $ax^2 + by^2 + cz^2 + 2hxy + 2gyz + 2fzx$   
 (iii)  $x_1^2 + x_2^2 + 3x_3^2 + x_4^2$   
 are quadratic forms in 2, 3 and 4 variables respectively.

**Definition 1.5.2** The general quadratic form in  $n$  variables  $x_1, x_2, \dots, x_n$  is

$$\sum_{j=1}^n \sum_{i=1}^n a_{ij} x_i x_j, \text{ where } a_{ij} \text{ are real numbers such that } a_{ij} = a_{ji} \text{ for all } i, j = 1, 2, 3, \dots, n.$$

Usually the quadratic form is denoted by  $Q$  and so  $Q = \sum_{j=1}^n \sum_{i=1}^n a_{ij} x_i x_j$

### 1. Matrix form of Q

If  $X = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}$ ,  $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$ , where  $a_{ij} = a_{ji}$ , then

$A$  is a symmetric matrix and the quadratic form

$$Q = \sum_{j=1}^n \sum_{i=1}^n a_{ij} x_i x_j \text{ can be written as } Q = X^T A X.$$

$A$  is called the matrix of the quadratic form.

For example the quadratic form  $x^2 + 4xy + 4y^2$  can be written in the matrix form

$$[x \ y] \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}. \text{ Here } X = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

**Note** Since the quadratic form  $Q = X^T A X$ , it is obvious that the characteristics or properties of  $Q$  depend on the characteristics of  $A$ .

### 2. Canonical form of Q

**Definition 1.5.3** A quadratic form  $Q$  which contains only the square terms of the variables is said to be in canonical form.

For example  $x^2 + y^2, x^2 - y^2, x^2 + y^2 - 4z^2$  and  $x_1^2 + x_2^2 + 2x_3^2 + x_4^2$  are in canonical forms because they contain only square terms.

### 3. Reduction of Q to canonical form by orthogonal transformation

Let  $Q = X^T A X$  be a quadratic form in  $n$  variables  $x_1, x_2, \dots, x_n$  and  $A = [a_{ij}]$  be the symmetric matrix of order  $n$  of the quadratic form. We will reduce  $A$  to diagonal form by an orthogonal transformation  $X = NY$ , where  $N$  is the normalised modal matrix of  $A$ . Then  $N^T A N = D$ , where  $D$  is the diagonal matrix containing the eigen values of  $A$ .

If  $\lambda_1, \lambda_2, \dots, \lambda_n$  are the eigen values of  $A$ , then

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

$\therefore$

$$Q = X^T A X = (NY)^T A (NY) \\ = Y^T (N^T A N) Y = Y^T D Y$$

$$\text{If } Y = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{bmatrix}, \text{ then } Q = [y_1 \ y_2 \ \dots \ y_n] \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{bmatrix} \\ = [\lambda_1 y_1 \ \lambda_2 y_2 \ \dots \ \lambda_n y_n] \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{bmatrix} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$$

**This is the required quadratic form.**

**Note** In the canonical form the coefficients are the eigen values of A. Since A is a symmetric matrix, the eigen values of A are all real. So, the eigen values may be positive, negative or zero. Hence the terms of the canonical form may be positive, negative or zero. By using the canonical form or the eigen values we can characterise the quadratic form.

**Definition 1.5.4** If A is the matrix of the quadratic form Q in the variables  $x_1, x_2, \dots, x_n$ , then the rank of Q is equal to the rank of A.

If rank of A  $< n$ , where  $n$  is the number of variables or order of A, then  $|A| = 0$  and Q is called a singular form.

#### 4. Index, signature and rank of quadratic form

**Definition 1.5.5** Let  $Q = X^T A X$  be a quadratic form in  $n$  variables  $x_1, x_2, \dots, x_n$ .

$X = [x_1, x_2, \dots, x_n]^T$  and A is the matrix of the quadratic form.

- (i) Index of the quadratic form is the number of positive eigen values of A.
- (ii) Signature of the quadratic form is the difference between the number of positive and negative eigen values of A.
- (iii) Rank of the quadratic form is the number of positive and negative eigen values of A.

Usually index is denoted by  $p$ , signature by  $s$  and rank by  $r$ .

**5. Definite and indefinite quadratic forms**

**Definition 1.5.6** Let  $Q = X^T A X$  be a quadratic form in  $n$  variables  $x_1, x_2, \dots, x_n$ .  
i.e.,  $X = [x_1 \dots x_n]^T$  and  $A$  is the matrix of the quadratic form.

- (i)  $Q$  is said to be **positive definite** if all the  $n$  eigen values of  $A$  are positive.  
i.e., if  $r = n$  and  $p = n$   
e.g.,  $y_1^2 + y_2^2 + \dots + y_n^2$  is positive definite.
  - (ii)  $Q$  is said to be **negative definite** if all the  $n$  eigen values of  $A$  are negative.  
i.e., if  $r = n, p = 0$   
e.g.,  $-y_1^2 - y_2^2 - \dots - y_n^2$  is negative definite.
  - (iii)  $Q$  is said to be **positive semidefinite** if all the  $n$  eigen values of  $A$  are  $\geq 0$  with at least one eigen value  $= 0$ .  
i.e., if  $r < n$  and  $p = r$   
e.g.,  $y_1^2 + y_2^2 + \dots + y_r^2$ , where  $r < n$ , is positive semi-definite.
  - (iv)  $Q$  is said to be **negative semidefinite** if all the  $n$  eigen values of  $A$  are  $\leq 0$  with at least one value  $= 0$ .  
i.e., if  $r < n$  and  $p = 0$   
e.g.,  $-y_1^2 - y_2^2 - \dots - y_r^2$ , where  $r < n$ , is negative semi definite.
  - (v)  $Q$  is said to be **indefinite** if  $A$  has positive and negative eigen values.  
e.g.,  $y_1^2 + y_2^2 - y_3^2 - y_4^2 + \dots + y_n^2$  is indefinite.
6. We can also find the nature of a quadratic form without finding the eigen values of  $A$  or without reducing to canonical form but by using the principal minors of  $A$  as below.

**Definition 1.5.7** Let  $Q = X^T A X$  be a quadratic form in  $n$  variables  $x_1, x_2, \dots, x_n$  and let the matrix of the quadratic form

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

$$\text{Let } D_1 = |a_{11}| = a_{11}, \quad D_2 = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix},$$

$$D_3 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \text{ and so on. Finally } D_n = |A|.$$

The determinants  $D_1, D_2, D_3, \dots, D_n$  are called the principal minors of  $A$ . The quadratic form  $Q$  is said to be



- (i) positive definite if  $D_i > 0$  for all  $i = 1, 2, \dots, n$
- (ii) negative definite if  $(-1)^i D_i > 0$  for all  $i = 1, 2, \dots, n$   
i.e.,  $D_1, D_3, D_5, \dots$  are negative and  $D_2, D_4, D_6, \dots$  are positive.
- (iii) positive semi-definite if  $D_i \geq 0$  for all  $i = 1, 2, 3, \dots, n$  and at least one  $D_i = 0$ .
- (iv) negative semi-definite if  $(-1)^i D_i \geq 0$  for all  $i = 1, 2, 3, \dots, n$  and at least one  $D_i = 0$ .
- (v) indefinite in all other cases.

### 7. Law of inertia of a quadratic form

In the reduction of a quadratic form to canonical form the number of positive and negative terms are independent of the choice of the transformation.

In other words, the signature of a real quadratic form is invariant under a real non-singular transformation. This property is called the law of inertia of the quadratic form.

## WORKED EXAMPLES

### Example 5

Write down the matrix of the quadratic form

$$2x_1^2 - 2x_2^2 + 4x_3^2 + 2x_1x_2 - 6x_1x_3 + 6x_2x_3.$$

[AU 2002]

### Solution

The Q.F is  $2x_1^2 - 2x_2^2 + 4x_3^2 + 2x_1x_2 - 6x_1x_3 + 6x_2x_3$

It has 3 variables  $x_1, x_2, x_3$ .

So, the matrix of the quadratic form is a  $3 \times 3$  symmetric matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Here

$$\begin{aligned} a_{11} &= \text{coefficient of } x_1^2 = 2 & a_{12} = a_{21} &= \frac{1}{2} (\text{coefficient of } x_1x_2) \\ & & &= \frac{1}{2} (2) = 1 \end{aligned}$$

$$\begin{aligned} a_{22} &= \text{coefficient of } x_2^2 = -2 & a_{13} = a_{31} &= \frac{1}{2} (\text{coefficient of } x_1x_3) \\ & & &= \frac{1}{2} (-6) = -3 \end{aligned}$$

$$a_{33} = \text{coefficient of } x_3^2 = 4$$

$$a_{23} = a_{32} = \frac{1}{2} (\text{coefficient of } x_2 x_3)$$

$$= \frac{1}{2} (6) = 3$$

$$\therefore A = \begin{bmatrix} 2 & 1 & -3 \\ 1 & -2 & 3 \\ -3 & 3 & 4 \end{bmatrix}$$

**Example 6**

Write down the quadratic form corresponding to the matrix

$$\begin{bmatrix} 2 & 4 & 5 \\ 4 & 3 & 1 \\ 5 & 1 & 1 \end{bmatrix}.$$

**Solution**

Let  $A = \begin{bmatrix} 2 & 4 & 5 \\ 4 & 3 & 1 \\ 5 & 1 & 1 \end{bmatrix}$ , which is a  $3 \times 3$  symmetric matrix.

So the quadratic form has 3 variables  $x_1, x_2, x_3$ .

Let  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ , then the quadratic form is

$$\begin{aligned} Q &= X^T A X = [x_1 \ x_2 \ x_3] \begin{bmatrix} 2 & 4 & 5 \\ 4 & 3 & 1 \\ 5 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= [2x_1 + 4x_2 + 5x_3 \ 4x_1 + 3x_2 + x_3 \ 5x_1 + x_2 + x_3] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= (2x_1 + 4x_2 + 5x_3)x_1 + (4x_1 + 3x_2 + x_3)x_2 + (5x_1 + x_2 + x_3)x_3 \\ &= 2x_1^2 + 4x_1x_2 + 5x_3x_1 + 4x_1x_2 + 3x_2^2 + x_3x_2 + 5x_1x_3 + x_2x_3 + x_3^2 \\ &= 2x_1^2 + 3x_2^2 + x_3^2 + 8x_1x_2 + 10x_1x_3 + 2x_2x_3 \end{aligned}$$

**Aliter** Given  $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 2 & 4 & 5 \\ 4 & 3 & 1 \\ 5 & 1 & 1 \end{bmatrix}$

Then the quadratic form in 3 variables  $x_1, x_2, x_3$  is  $Q = X^T A X$

$$\begin{aligned} &= a_{11}x_1^2 + a_{22}x_2^2 + a_{33}x_3^2 + 2(a_{12})x_1x_2 + 2(a_{13})x_1x_3 + 2(a_{23})x_2x_3 \\ &= 2x_1^2 + 3x_2^2 + x_3^2 + 2(4)x_1x_2 + 2(5)x_1x_3 + 2(1)x_2x_3 \\ &= 2x_1^2 + 3x_2^2 + x_3^2 + 8x_1x_2 + 10x_1x_3 + 2x_2x_3 \end{aligned}$$

**Example 7****Determine the nature of the following quadratic form**

$$f(x_1, x_2, x_3) = x_1^2 + 2x_2^2$$

**[AU 2003]****Solution**

The given quadratic form in 3 variables  $x_1, x_2, x_3$  contains only square terms and so it is in canonical form.

The coefficients are the eigen values. Here  $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 0$ , since there is no  $x_3^2$  term.

$\therefore$  It is a positive semi-definite form.

**Example 8****Find the nature of the quadratic form  $2x^2 + 2xy + 3y^2$ .****Solution**

The given quadratic form is  $2x^2 + 2xy + 3y^2$  having 2 variables. So, the matrix of the quadratic form is  $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ .

The principal minors are  $D_1 = 2 > 0$  and  $D_2 = \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = 5 > 0$

$\therefore$  the quadratic form is positive definite.

**Example 9****Discuss the nature of the following quadratic forms.**

(i)  $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4zx$

(ii)  $6x_1^2 + 3x_2^2 + 14x_3^2 + 4x_2x_3 + 18x_1x_3 + 4x_1x_2$

(iii)  $xy + yz + zx$

(iv)  $10x^2 + 2y^2 + 5z^2 + 6yz - 10zx - 4xy$ .

**Solution**

(i) The Q.F is  $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4zx$ , having 3 variables  $x, y, z$ .

The matrix of the quadratic form is

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}.$$

The principal minors are

$$D_1 = |6| = 6 > 0; D_2 = \begin{vmatrix} 6 & -2 \\ -2 & 3 \end{vmatrix} = 18 - 4 = 14 > 0$$

$$D_3 = |A| = \begin{vmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{vmatrix}$$

$$= 6(9 - 1) + 2(-6 + 2) + 2(2 - 6) = 48 - 8 - 8 = 32 > 0$$

Since  $D_1, D_2, D_3$  are positive, the quadratic form is positive definite.

- (ii) The quadratic form is  $6x_1^2 + 3x_2^2 + 14x_3^2 + 4x_2x_3 + 18x_1x_3 + 4x_1x_2$ , having 3 variables  $x_1, x_2, x_3$ . The matrix of the Q.F is

$$A = \begin{bmatrix} 6 & 2 & 9 \\ 2 & 3 & 2 \\ 9 & 2 & 14 \end{bmatrix}.$$

The principal minors are

$$D_1 = 6 > 0; D_2 = \begin{vmatrix} 6 & 2 \\ 2 & 3 \end{vmatrix} = 18 - 4 = 14 > 0$$

$$\begin{aligned} D_3 = |A| &= \begin{vmatrix} 6 & 2 & 9 \\ 2 & 3 & 2 \\ 9 & 2 & 14 \end{vmatrix} \\ &= 6(42 - 4) - 2(28 - 18) + 9(4 - 27) \\ &= 6(38) - 20 + 9(-23) \\ &= 228 - 20 - 207 = 1 > 0 \end{aligned}$$

Since  $D_1, D_2, D_3$  are positive, the quadratic form is positive definite.

- (iii) The quadratic form is  $xy + yz + zx$  in 3 variables  $x, y, z$

$$\text{The matrix of the quadratic form is } A = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

$$\text{The principal minors are } D_1 = 0, D_2 = \begin{vmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{vmatrix} = -\frac{1}{4} < 0$$

$$\begin{aligned} D_3 &= \begin{vmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{vmatrix} \\ &= \frac{1}{8} \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = \frac{1}{8} [0 - 1(0 - 1) + (1 - 0)] \\ &= \frac{1}{8} \times 2 = \frac{1}{4} > 0 \end{aligned}$$

Since  $D_1 = 0, D_2 < 0, D_3 > 0$ , the quadratic form is indefinite.

- (iv) The Q.F  $10x^2 + 2y^2 + 5z^2 + 6yz - 10zx - 4xy$  is in three variables  $x, y, z$ .

The matrix of the quadratic form  $A = \begin{bmatrix} 10 & -2 & -5 \\ -2 & 2 & 3 \\ -5 & 3 & 5 \end{bmatrix}$

The principal minors are  $D_1 = 10 > 0$

$$D_2 = \begin{vmatrix} 10 & -2 \\ -2 & 2 \end{vmatrix} = 20 - 4 = 16 > 0$$

$$D_3 = \begin{vmatrix} 10 & -2 & -5 \\ -2 & 2 & 3 \\ -5 & 3 & 5 \end{vmatrix} = 10(10 - 9) + 2(-10 + 15) - 5(-6 + 10) \\ = 10 + 10 - 20 = 0$$

Since  $D_1 > 0$ ,  $D_2 > 0$  and  $D_3 = 0$ , the quadratic form is positive semi-definite.

### Example 10

Determine  $\lambda$  so that  $\lambda(x^2 + y^2 + z^2) + 2xy - 2yz + 2zx$  is positive definite.

### Solution

The matrix of the quadratic form is  $A = \begin{bmatrix} \lambda & 1 & 1 \\ 1 & \lambda & -1 \\ 1 & -1 & \lambda \end{bmatrix}$

The principal minors are  $D_1 = \lambda$

$$D_2 = \begin{vmatrix} \lambda & 1 \\ 1 & \lambda \end{vmatrix} = \lambda^2 - 1; D_3 = |A| = \lambda(\lambda^2 - 1) - (\lambda + 1) + (-1 - \lambda) \\ = \lambda^3 - 3\lambda - 2 = (\lambda + 1)^2(\lambda - 2)$$

Given the quadratic form is positive definite.

$$\therefore D_1 > 0, D_2 > 0 \text{ and } D_3 > 0$$

$$\Rightarrow \lambda > 0, \lambda^2 - 1 > 0 \Rightarrow (\lambda + 1)(\lambda - 1) > 0$$

$$\Rightarrow \lambda < -1 \text{ or } \lambda > 1$$

$$\text{and } (\lambda + 1)^2(\lambda - 2) > 0 \Rightarrow \lambda - 2 > 0 \quad [\because (\lambda + 1)^2 > 0]$$

$$\Rightarrow \lambda > 2$$

$$\therefore \text{ the common values of } \lambda \text{ are } \lambda > 2$$

### Example 11

Show that the quadratic form  $ax_1^2 - 2bx_1x_2 + cx_2^2$  is positive definite if  $a > 0$  and  $ac - b^2 > 0$ .

### Solution

The matrix of the quadratic form is  $A = \begin{bmatrix} a & -b \\ -b & c \end{bmatrix}$

The principal minors are  $D_1 = a$ ,  $D_2 = |A| = \begin{vmatrix} a & -b \\ -b & c \end{vmatrix} = ac - b^2$ .

Given  $a > 0$  and  $ac - b^2 > 0$ .

$\therefore D_1 > 0$  and  $D_2 > 0$ .

Hence the Q.F is positive definite.

### Example 12

**Find the index, signature and rank of the Q.F in 3 variables  $x_1^2 + 2x_2^2 - 3x_3^2$ .**

[AU 2006, 2008]

### Solution

The quadratic form is in canonical form in 3 variables.

$\therefore \lambda_1 = 1, \lambda_2 = 2, \lambda_3 = -3$ .

There are 2 positive eigen values.  $\therefore$  Index = 2

One negative eigen value  $\therefore$  signature = 2 - 1 = 1

Rank = 3 (i.e., the total number of +ve and -ve eigen values)

**Note** Its nature is indefinite.

### Example 13

**Reduce  $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$  into a canonical form by an orthogonal reduction and find the rank, signature, index and the nature of the quadratic form.**

[AU 1996, 2005, 2006, 2010]

### Solution

Given quadratic form is  $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$

The matrix of the Q.F is  $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

The characteristic equation is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1 = 6 + 3 + 3 = 12$

$$S_2 = \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} + \begin{vmatrix} 6 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 6 & -2 \\ -2 & 3 \end{vmatrix} = 8 + 14 + 14 = 36$$

$$S_3 = |A| = 6(9 - 1) + 2(-6 + 2) + 2(2 - 6) = 48 - 8 - 8 = 32$$

$\therefore$  the characteristic equation is  $\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$

By trial,  $\lambda = 2$  is a root.

Other roots are given by  $\lambda^2 - 10\lambda + 16 = 0$

$$\Rightarrow (\lambda - 2)(\lambda - 8) = 0$$

$$\Rightarrow \lambda = 2, 8$$

$\therefore$  the eigen values are  $\lambda = 2, 2, 8$

$$2 \left| \begin{array}{ccc|c} 1 & -12 & 36 & -32 \\ 0 & 2 & -20 & 32 \\ 1 & -10 & 16 & 0 \end{array} \right|$$

**To find eigen vectors:**

Let  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector corresponding to eigen value  $\lambda$ .

$$\text{Then } (A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left. \begin{aligned} (6-\lambda)x_1 - 2x_2 + 2x_3 &= 0 \\ -2x_1 + (3-\lambda)x_2 - x_3 &= 0 \\ 2x_1 - x_2 + (3-\lambda)x_3 &= 0 \end{aligned} \right\} \quad \text{(I)}$$

**Case (i)** If  $\lambda = 8$ , then equations (I) become

$$-2x_1 - 2x_2 + 2x_3 = 0 \Rightarrow x_1 + x_2 - x_3 = 0$$

$$-2x_1 - 5x_2 - x_3 = 0 \Rightarrow 2x_1 + 5x_2 + x_3 = 0$$

$$2x_1 - x_2 - 5x_3 = 0$$

From the first two equations we get

$$\Rightarrow \frac{x_1}{1+5} = \frac{x_2}{-2-1} = \frac{x_3}{5-2}$$

$$\Rightarrow \frac{x_1}{6} = \frac{x_2}{-3} = \frac{x_3}{3}$$

$$\Rightarrow \frac{x_1}{2} = \frac{x_2}{-1} = \frac{x_3}{1}$$

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ 1 & -1 & 1 \\ 5 & 1 & 2 \end{array} \quad \begin{array}{ccc} 1 & -1 & 1 \\ 5 & 1 & 2 \end{array}$$

Choosing  $x_1 = 2, x_2 = -1, x_3 = 1$ , we get an eigen vector  $X_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

**Case (ii)** If  $\lambda = 2$ , then equations (I) become

$$4x_1 - 2x_2 + 2x_3 = 0 \Rightarrow 2x_1 - x_2 + x_3 = 0$$

$$-2x_1 + x_2 - x_3 = 0 \Rightarrow 2x_1 - x_2 + x_3 = 0$$

$$2x_1 - x_2 + x_3 = 0$$

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So we get only one equation  $2x_1 - x_2 + x_3 = 0$

Choosing  $x_3 = 0$ , we get  $2x_1 - 2x_2 = 0 \Rightarrow x_2 = 2x_1$

Choosing  $x_1 = 1$ , we get  $x_2 = 2$

$\therefore$  an eigen vector is 
$$X_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

We shall find another eigen vector  $X_3 = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$  orthogonal to  $X_2$

$\therefore$   $a + 2b = 0 \Rightarrow a = -2b$

Also  $X_3$  satisfies (1)

$\therefore$   $2a - b + c = 0$

$\Rightarrow -4b - b + c = 0$

$\Rightarrow c = 5b$

Choosing  $b = 1$  we get  $c = 5$  and  $a = -2$

$\therefore$  
$$X_3 = \begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix}$$

Thus the eigen values are 8, 2, 2 and the corresponding eigen vectors are

$$X_1 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad X_3 = \begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix}$$

Clearly  $X_3$  is orthogonal to  $X_1$  and  $X_2$ .

But  $X_1$  and  $X_2$  are also orthogonal.

The normalised eigen vectors are 
$$\begin{bmatrix} \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}, \quad \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \\ 0 \end{bmatrix}, \quad \begin{bmatrix} \frac{-2}{\sqrt{30}} \\ \frac{1}{\sqrt{30}} \\ \frac{5}{\sqrt{30}} \end{bmatrix}$$

The normalised modal matrix  $N = \begin{bmatrix} \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{5}} & \frac{-2}{\sqrt{30}} \\ \frac{-1}{\sqrt{6}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{30}} \\ \frac{1}{\sqrt{6}} & 0 & \frac{5}{\sqrt{30}} \end{bmatrix}$



$$\text{Then } N^T A N = D = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

The orthogonal transformation  $X = NY$ , where  $Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$  reduces the given

$$\text{quadratic form to } Y^T D Y = [y_1 \ y_2 \ y_3] \begin{bmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$= 8y_1^2 + 2y_2^2 + 2y_3^2, \text{ which is the canonical form.}$$

$\therefore$  rank of the Q.F = 3

index = 3

signature = 3

The Q.F is positive definite since all the eigen values are positive.

### Example 14

**Reduce the quadratic form  $x_1^2 + 5x_2^2 + x_3^2 + 2x_1x_2 + 2x_2x_3 + 6x_3x_1$  to canonical form through an orthogonal transformation.**

[AU 2003, 2006, 2010]

### Solution

$$\text{The matrix of the quadratic form is } A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

$$\text{where } S_1 = 1 + 5 + 1 = 7$$

$$S_2 = \begin{vmatrix} 5 & 1 \\ 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix} = 4 + (-8) + 4 = 0$$

$$S_3 = |A| = 1 \cdot 4 - (-2) + 3(-14) = 4 + 2 - 42 = -36$$

$$\therefore \text{ the characteristic equation is } \lambda^3 - 7\lambda^2 + 36 = 0$$

By trial  $\lambda = -2$  is a root.

Other roots are given by

$$\begin{aligned} & \lambda^2 - 9\lambda + 18 = 0 \\ \Rightarrow & (\lambda - 6)(\lambda - 3) = 0 \\ \Rightarrow & \lambda = 6, +3 \end{aligned}$$

$\therefore$  the eigen values are  $\lambda = -2, 3, 6$

$$-2 \quad \left| \begin{array}{ccc|c} 1 & -7 & 0 & 36 \\ 0 & -2 & 18 & -36 \\ 1 & -9 & 18 & 0 \end{array} \right|$$

**To find eigen vectors:**

Let  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector corresponding to eigen value  $\lambda$ .

$$\text{Then } (A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left. \begin{aligned} (1-\lambda)x_1 + x_2 + 3x_3 &= 0 \\ x_1 + (5-\lambda)x_2 + x_3 &= 0 \\ 3x_1 + x_2 + (1-\lambda)x_3 &= 0 \end{aligned} \right\} \quad \text{(I)}$$

**Case (i)** If  $\lambda = -2$ , then the equations (I) become

$$3x_1 + x_2 + 3x_3 = 0$$

$$x_1 + 7x_2 + x_3 = 0$$

$$3x_1 + x_2 + 3x_3 = 0$$

From first two equations, we get

$$\begin{aligned} \Rightarrow \quad \frac{x_1}{1-21} &= \frac{x_2}{3-3} = \frac{x_3}{21-1} & \begin{array}{ccc} x_1 & x_2 & x_3 \\ 1 & 3 & 3 \\ 7 & 1 & 1 \end{array} \\ \Rightarrow \quad \frac{x_1}{-20} &= \frac{x_2}{0} = \frac{x_3}{20} & \begin{array}{ccc} & & \\ & \times & \\ & & \end{array} \\ \Rightarrow \quad \frac{x_1}{1} &= \frac{x_2}{0} = \frac{x_3}{-1} & \begin{array}{ccc} & & \\ & \times & \\ & & \end{array} \end{aligned}$$

Choosing  $x_1 = 1, x_2 = 0, x_3 = -1$ , we get an eigen vector  $X_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

**Case (ii)** If  $\lambda = 3$ , then equations (I) become

$$-2x_1 + x_2 + 3x_3 = 0$$

$$x_1 + 2x_2 + x_3 = 0$$

$$3x_1 + x_2 - 2x_3 = 0$$

From the first two equations, we get

$$\begin{aligned} \frac{x_1}{1-6} &= \frac{x_2}{3+2} = \frac{x_3}{-4-1} \\ \Rightarrow \frac{x_1}{-5} &= \frac{x_2}{5} = \frac{x_3}{-5} \\ \Rightarrow \frac{x_1}{1} &= \frac{x_2}{-1} = \frac{x_3}{1} \end{aligned}$$

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ 1 & 3 & -2 \\ 2 & 1 & 1 \end{array} \quad \begin{array}{ccc} 1 & 3 & -2 \\ 2 & 1 & 1 \end{array}$$

Choosing  $x_1 = 1, x_2 = -1, x_3 = 1$  we get an eigen vector  $X_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ .

**Case (iii)** If  $\lambda = 6$ , then equations (I) become

$$\begin{aligned} -5x_1 + x_2 + 3x_3 &= 0 \\ x_1 - x_2 + x_3 &= 0 \\ 3x_1 + x_2 - 5x_3 &= 0 \end{aligned}$$

From the first two equations, we get

$$\begin{aligned} \frac{x_1}{1+3} &= \frac{x_2}{3+5} = \frac{x_3}{5-1} \\ \Rightarrow \frac{x_1}{4} &= \frac{x_2}{8} = \frac{x_3}{4} \\ \Rightarrow \frac{x_1}{1} &= \frac{x_2}{2} = \frac{x_3}{1} \end{aligned}$$

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ 1 & 3 & -5 \\ -1 & 1 & 1 \end{array} \quad \begin{array}{ccc} 1 & 3 & -5 \\ -1 & 1 & 1 \end{array}$$

Choosing  $x_1 = 1, x_2 = 2, x_3 = 1$  we get an eigen vector  $X_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ .

Since the eigen values are different, the eigen vectors are mutually orthogonal.

Normalised eigen vectors are  $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$

The normalised modal matrix  $N = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix}$

Then

$$\begin{aligned}
 N^T A N &= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix} \\
 &= \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix} = D
 \end{aligned}$$

The transformation  $X = NY$ , where  $Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$  reduces the quadratic form into  $Y^T$

$$\begin{aligned}
 DY &= [y_1 \ y_2 \ y_3] \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\
 &= -2y_1^2 + 3y_2^2 + 6y_3^2 \text{ which is the canonical form.}
 \end{aligned}$$

### Example 15

**Find out the type of conic represented by  $17x^2 - 30xy + 17y^2 = 128$  after reducing the quadratic form  $17x^2 - 30xy + 17y^2$  to canonical form by an orthogonal transformation.**

### Solution

Given quadratic form is  $17x^2 - 30xy + 17y^2$

The matrix of the Q.F is  $A = \begin{bmatrix} 17 & -15 \\ -15 & 17 \end{bmatrix}$

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 17 - \lambda & -15 \\ -15 & 17 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (17 - \lambda)^2 - 15^2 = 0$$

$$\Rightarrow (17 - \lambda)^2 = 15^2$$

$$\Rightarrow 17 - \lambda = \pm 15$$

$$\Rightarrow \lambda = 2 \text{ or } 32.$$

**To find eigen vectors:**

If  $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$  be an eigen vector corresponding to eigen value  $\lambda$ .

$$\begin{aligned}
 \text{Then } (A - \lambda I)X = 0 &\Rightarrow \begin{bmatrix} 17 - \lambda & -15 \\ -15 & 17 - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\
 \Rightarrow &\quad \left. \begin{aligned} (17 - \lambda)x_1 - 15x_2 &= 0 \\ -15x_1 + (17 - \lambda)x_2 &= 0 \end{aligned} \right\} \quad (I)
 \end{aligned}$$

**Case (i)** If  $\lambda = 2$ , then equations (I) become

$$\begin{aligned}
 15x_1 - 15x_2 &= 0 \\
 -15x_1 + 15x_2 &= 0 \Rightarrow x_1 = x_2
 \end{aligned}$$

Choosing  $x_1 = 1$ , we get  $x_2 = 1$

$$\therefore \text{ an eigen vector is } X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

**Case (ii)** If  $\lambda = 32$ , then equation (I) become

$$\begin{aligned}
 -15x_1 - 15x_2 &= 0 \\
 -15x_1 - 15x_2 &= 0 \Rightarrow x_2 = -x_1
 \end{aligned}$$

Choose  $x_1 = 1$ , we get  $x_2 = -1$

$$\therefore \text{ an eigen vector is } X_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\therefore \text{ the normalised eigen vectors are } \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

The normalised modal matrix

$$\begin{aligned}
 N &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \\
 \Rightarrow N^T A N &= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 17 & -15 \\ -15 & 17 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \\
 &= \begin{bmatrix} 2 & 0 \\ 0 & 32 \end{bmatrix} = D
 \end{aligned}$$

The transformation  $X = NY$ , where  $Y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$  reduces the given quadratic form to

$$Y^T D Y = [y_1 y_2] \begin{bmatrix} 2 & 0 \\ 0 & 32 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$= 2y_1^2 + 32y_2^2, \text{ which is the canonical form.}$$

But the given Q.F = 128

$$\therefore 2y_1^2 + 32y_2^2 = 128$$

$$\Rightarrow \frac{y_1^2}{64} + \frac{y_2^2}{4} = 1, \text{ which represents an ellipse.}$$

### Example 16

**Reduce the quadratic form  $8x^2 + 7y^2 + 3z^2 - 12xy + 4xz - 8yz$  to the canonical form by an orthogonal transformation. Find one set of values of  $x, y, z$  (not all zero) which will make the quadratic form zero.**

[AU 2002, 2009]

### Solution

Given quadratic form is  $8x^2 + 7y^2 + 3z^2 - 12xy + 4xz - 8yz$

The matrix of the quadratic form is  $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$

The characteristic equation is  $|A - \lambda I| = 0$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1 = 8 + 7 + 3 = 18$

$$S_2 = \begin{vmatrix} 7 & -4 \\ -4 & 3 \end{vmatrix} + \begin{vmatrix} 8 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix}$$

$$= (21 - 16) + (24 - 4) + (56 - 36)$$

$$= 5 + 20 + 20 = 45$$

$$S_3 = |A| = 8(21 - 16) + 6(-18 + 8) + 2(24 - 14)$$

$$= 40 - 60 + 20 = 0$$

$\therefore$  the characteristic equation is  $\lambda^3 - 18\lambda^2 + 45\lambda = 0$

$$\Rightarrow \lambda(\lambda^2 - 18\lambda + 45) = 0$$

$$\Rightarrow \lambda(\lambda - 3)(\lambda - 15) = 0$$

$$\Rightarrow \lambda = 0, 3, 15$$

**To find eigen vectors:**

If  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector of the eigen value  $\lambda$  of A, then

$$(A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \left. \begin{aligned} (8-\lambda)x_1 - 6x_2 + 2x_3 &= 0 \\ -6x_1 + (7-\lambda)x_2 - 4x_3 &= 0 \\ 2x_1 - 4x_2 + (3-\lambda)x_3 &= 0 \end{aligned} \right\} \quad (I)$$

**Case (i)** If  $\lambda = 0$ , then the equations (I) become

$$\begin{aligned} 8x_1 - 6x_2 + 2x_3 &= 0 \Rightarrow 4x_1 - 3x_2 + x_3 = 0 \\ -6x_1 + 7x_2 - 4x_3 &= 0 \\ 2x_1 - 4x_2 + 3x_3 &= 0 \end{aligned}$$

From the first and third equations we get

$$\begin{aligned} \Rightarrow \frac{x_1}{-9+4} &= \frac{x_2}{2-12} = \frac{x_3}{-16+6} & \begin{array}{ccc} x_1 & x_2 & x_3 \\ -3 & 1 & 4 \\ -4 & 3 & 2 \end{array} \\ \Rightarrow \frac{x_1}{-5} &= \frac{x_2}{-10} = \frac{x_3}{-10} \\ \Rightarrow \frac{x_1}{1} &= \frac{x_2}{2} = \frac{x_3}{2} \end{aligned}$$

Choosing  $x_1 = 1, x_2 = 2, x_3 = 2$ , we get an eigen vector

$$X_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Similarly, we can find for  $\lambda = 3, X_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$  and for  $\lambda = 15, X_3 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$

$$\begin{aligned} \therefore \text{the normalised modal matrix } N &= \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{-2}{3} \\ \frac{2}{3} & \frac{-2}{3} & \frac{1}{3} \end{bmatrix} \\ &= \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix} \end{aligned}$$

$$\therefore N^T A N = D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix}$$

The transformation  $X = NY$ , where  $Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ , reduces the quadratic form to the canonical form

$$\begin{aligned} Y^T D Y &= [y_1 y_2 y_3] \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\ &= 0y_1^2 + 3y_2^2 + 15y_3^2 \end{aligned}$$

The transformation is  $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$

$$\therefore x = \frac{1}{3}(y_1 + 2y_2 + 2y_3)$$

$$y = \frac{1}{3}(2y_1 + y_2 - 2y_3)$$

$$z = \frac{1}{3}(2y_1 - 2y_2 + y_3)$$

Quadratic form = 0

$$\Rightarrow 0y_1^2 + 3y_2^2 + 15y_3^2 = 0$$

$$\Rightarrow 3y_2^2 + 15y_3^2 = 0$$

$$\Rightarrow y_2 = 0 \text{ and } y_3 = 0$$

and  $y_1$  can take any value, we shall choose  $y_1 = 3$

$$\therefore x = 1, y = 2, z = 2$$

Hence, this set of values will make the quadratic form = 0

### Example 17

**Reduce the quadratic form  $2x_1^2 + 6x_2^2 + 2x_3^2 + 8x_1x_3$  to canonical form.**

[AU 2008, 2010]

### Solution

The quadratic form is  $2x_1^2 + 6x_2^2 + 2x_3^2 + 8x_1x_3$

$$\therefore \text{the matrix of the quadratic form is } A = \begin{bmatrix} 2 & 0 & 4 \\ 0 & 6 & 0 \\ 4 & 0 & 2 \end{bmatrix}$$



The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 0 & 4 \\ 0 & 6-\lambda & 0 \\ 4 & 0 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1 = 2 + 6 + 2 = 10$

$$S_2 = \begin{vmatrix} 6 & 0 \\ 0 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 4 \\ 4 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 0 \\ 0 & 6 \end{vmatrix}$$

$$= 12 + (4 - 16) + 12 = 12$$

$$S_3 = |A| = 2(12) + 4(-24) = -72$$

$\therefore$  Characteristic equation is  $\lambda^3 - 10\lambda^2 + 12\lambda + 72 = 0$

By trial we find  $\lambda = -2$  is a root

Other roots are given by  $\lambda^2 - 12\lambda + 36 = 0$

$$\Rightarrow (\lambda - 6)^2 = 0$$

$$\Rightarrow \lambda = 6, 6$$

$$\begin{array}{ccc|c} 1 & -10 & 12 & 72 \\ 0 & -2 & 24 & -72 \\ 0 & -12 & 36 & 0 \end{array}$$

$\therefore$  the eigen values are  $\lambda = -2, 6, 6$

**To find eigen vectors:**

If  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector corresponding to the eigen value  $\lambda$  of A.

Then

$$(A - \lambda I)X = 0$$

$$\Rightarrow \begin{bmatrix} 2-\lambda & 0 & 4 \\ 0 & 6-\lambda & 0 \\ 4 & 0 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left. \begin{aligned} (2-\lambda)x_1 + 0x_2 + 4x_3 &= 0 \\ 0x_2 + (6-\lambda)x_2 + 0x_3 &= 0 \\ 4x_1 + 0x_2 + (2-\lambda)x_3 &= 0 \end{aligned} \right\} \quad (I)$$

**Case (i)** If  $\lambda = -2$ , then the equations (I) become

$$4x_1 + 4x_3 = 0 \Rightarrow x_1 + x_3 = 0 \Rightarrow x_3 = -x_1$$

$$8x_2 = 0 \Rightarrow x_2 = 0$$

Taking  $x_1 = 1$ , we get  $x_3 = -1$

$$\therefore \text{an eigen vector is } X_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

**Case (ii)** If  $\lambda = 6$ , then equations (I) become

$$-4x_1 + 4x_3 = 0 \Rightarrow x_1 = x_3$$

and  $0x_2 = 0 \Rightarrow x_2$  can take any value.

If  $x_1 = 1$ , then  $x_3 = 1$ . Choose  $x_2 = 0$

$\therefore$  an eigen vector is 
$$X_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

We shall choose an eigen vector  $X_3$  orthogonal to  $X_2$

If  $x_1 = 0$ ,  $x_3 = 0$ , choose  $x_2 = 1$ , then  $X_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$  and clearly  $X_2$  and  $X_3$  are orthogonal vectors.

The normalised eigen vectors are 
$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$\therefore$  The normalised modal matrix  $N = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$

Then 
$$N^T A N = D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

The transformation  $X = NY$ , where  $Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ , reduces the quadratic form to

canonical form  $Y^T D Y = -2y_1^2 + 6y_2^2 + 6y_3^2$

Since the canonical form has positive and negative terms, the quadratic form is indefinite in nature.

### Example 18

**Reduce the given quadratic form  $Q$  to its canonical form using orthogonal transformation  $Q = x^2 + 3y^2 + 3z^2 - 2yz$  [AU 2009]**

**Solution**

Given quadratic form is  $Q = x^2 + 3y^2 + 3z^2 - 2yz$

The matrix of the quadratic form is  $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$

The characteristic equation is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 3-\lambda & -1 \\ 0 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1$  = sum of the diagonal elements of A.

$$= 1 + 3 + 3 = 7$$

$S_2$  = sum of minors of elements of diagonals of A.

$$= \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 0 & 3 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 0 & 3 \end{vmatrix}$$

$$= 9 - 1 + 3 - 0 + (3 - 0) = 14$$

$$S_3 = |A| = 1(9 - 1) = 8$$

$\therefore$  the characteristic equation is  $\lambda^3 - 7\lambda^2 + 14\lambda - 8 = 0$

Since the sum of the coefficients is 0,

$\lambda = 1$  is a root.

The other roots are given by  $\lambda^2 - 6\lambda + 8 = 0$

$$\Rightarrow (\lambda - 2)(\lambda - 4) = 0 \Rightarrow \lambda = 2, 4$$

$\therefore$  the eigen values are  $\lambda = 1, 2, 4$

$$\begin{array}{ccc|ccc} 1 & 1 & -7 & 14 & -8 & \\ & 0 & 1 & -6 & 8 & \\ & 1 & -6 & 8 & 0 & \end{array}$$

**To find eigen vectors:**

Let  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector corresponding to an eigen value  $\lambda$  of A.

$$\text{Then } (A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} 1-\lambda & 0 & 0 \\ 0 & 3-\lambda & -1 \\ 0 & -1 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left. \begin{array}{l} (1-\lambda)x_1 + 0x_2 + 0x_3 = 0 \\ 0x_1 + (3-\lambda)x_2 + (-1)x_3 = 0 \\ 0x_1 + (-1)x_2 + (3-\lambda)x_3 = 0 \end{array} \right\} \quad (I)$$

**Case (i)** If  $\lambda = 1$ , then equations (I) become

$$0x_1 + 0x_2 + x_3 = 0 \Rightarrow x_3 = 0$$

$$0x_1 + 2x_2 - x_3 = 0 \Rightarrow 2x_2 = x_3$$

$$0x_1 - x_2 + 2x_3 = 0 \Rightarrow 2x_2 = 2x_3$$

$$\therefore x_3 = 0 \Rightarrow x_2 = 0$$

$x_1$  can be any real value. We shall take  $x_1 = 1$

$$\therefore \text{an eigen vector is } X_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

**Case (ii)** If  $\lambda = 2$ , then the equations (I) become

$$-x_1 = 0 \Rightarrow x_1 = 0$$

$$0x_1 + x_2 - x_3 = 0 \Rightarrow x_2 = x_3$$

$$0x_1 - x_2 + x_3 = 0 \Rightarrow x_2 = x_3$$

Take  $x_2 = 1 \therefore x_3 = 1$

$$\therefore \text{an eigen vector is } X_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

**Case (iii)** If  $\lambda = 4$ , then the equations (I) become

$$-3x_1 + 0x_2 + 0x_3 = 0 \Rightarrow x_1 = 0$$

$$0x_1 - x_2 - x_3 = 0 \Rightarrow x_2 = -x_3$$

$$0x_1 - x_2 - x_3 = 0 \Rightarrow x_2 = -x_3$$

Take  $x_3 = 1 \therefore x_2 = -1$

$$\therefore \text{an eigen vector is } X_3 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

Thus the eigen values are  $\lambda = 1, 2, 4$  and the corresponding eigen vectors are

$$X_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad X_3 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$\text{The normalized eigen vectors are } \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \begin{bmatrix} 0 \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

So, the normalized modal matrix  $N = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$

$$\therefore N^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\therefore AN = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{2}{\sqrt{2}} & -\frac{4}{\sqrt{2}} \\ 0 & \frac{2}{\sqrt{2}} & \frac{4}{\sqrt{2}} \end{bmatrix}$$

$$\therefore N^T AN = D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{2}{\sqrt{2}} & -\frac{4}{\sqrt{2}} \\ 0 & \frac{2}{\sqrt{2}} & \frac{4}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

The orthogonal transformation  $X = NY$ , where  $Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ , reduces the given

$$\text{quadratic form to } Y^T DY = [y_1 \ y_2 \ y_3] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$= y_1^2 + 2y_2^2 + 4y_3^2,$$

which is the canonical form.

**Example 19**

Reduce the quadratic form  $x_1^2 + 2x_2^2 + x_3^2 - 2x_1x_2 + 2x_2x_3$  to the canonical form through an orthogonal transformation and hence show that it is positive semi-definite. Also give a non-zero set of values  $(x_1, x_2, x_3)$  which makes the quadratic form zero. [AU 2008, 2009]

**Solution**

Given the quadratic form is  $x_1^2 + 2x_2^2 + x_3^2 - 2x_1x_2 + 2x_2x_3$

$$\text{The matrix of the quadratic form is } A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 1-\lambda & -1 & 0 \\ -1 & 2-\lambda & 1 \\ 0 & 1 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where  $S_1 = 1 + 2 + 1 = 4$

$$S_2 = \begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & -1 \\ -1 & 2 \end{vmatrix} = 1 + 1 + 1 = 3$$

$$S_3 = |A| = 2 - 1 + 1(-1) = 0$$

$\therefore$  the characteristic equation is  $\lambda^3 - 4\lambda^2 + 3\lambda = 0$

$$\Rightarrow \lambda(\lambda^2 - 4\lambda + 3) = 0$$

$$\Rightarrow \lambda(\lambda - 1)(\lambda - 3) = 0 \Rightarrow \lambda = 0, 1, 3$$

**To find eigen vectors:**

If  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be an eigen vector corresponding to an eigen value  $\lambda$  of A, then

$$(A - \lambda I)X = 0 \Rightarrow \begin{bmatrix} 1-\lambda & -1 & 0 \\ -1 & 2-\lambda & 1 \\ 0 & 1 & 1-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow (1-\lambda)x_1 - x_2 + 0x_3 = 0$$

$$-x_1 + (2-\lambda)x_2 + x_3 = 0$$

$$0x_1 + x_2 + (1-\lambda)x_3 = 0$$

(I)

**Case (i)** If  $\lambda = 0$ , then the equations (I) become

$$\begin{aligned}x_1 - x_2 &= 0 \Rightarrow x_1 = x_2 \\-x_1 + 2x_2 + x_3 &= 0 \\x_2 + x_3 &= 0 \Rightarrow x_3 = -x_2\end{aligned}$$

Take  $x_2 = 1$ , then  $x_1 = 1, x_3 = -1$

$\therefore$  an eigen vector is  $X_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

**Case (ii)** If  $\lambda = 1$ , the equations (I) become

$$\begin{aligned}0x_1 - x_2 &= 0 \Rightarrow x_2 = 0 \\-x_1 + x_2 + x_3 &= 0 \Rightarrow x_1 = x_3\end{aligned}$$

Take  $x_3 = 1 \quad \therefore$  an eigen vector is  $X_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

**Case (iii)** If  $\lambda = 3$ , then equations (I) become

$$\begin{aligned}-2x_1 - x_2 &= 0 \Rightarrow x_2 = -2x_1 \\-x_1 - x_2 + x_3 &= 0 \\x_2 - 2x_3 &= 0 \Rightarrow x_2 = 2x_3\end{aligned}$$

Take  $x_2 = 2$ , then  $x_1 = -1, x_3 = 1$

$\therefore$  an eigen vector is  $X_3 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$

Thus the eigen values are  $\lambda = 0, 1, 3$  and the eigen vectors are

$$X_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad X_3 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

$\therefore$  the normalized eigen vectors are  $\begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{-1}{\sqrt{3}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$

$$\therefore \text{ the normalised modal matrix is } N = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

$$\therefore \text{ the diagonal matrix is } N^T A N = D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

The orthogonal transformation is  $X = NY$

$$\text{where } Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \text{ and the canonical form is } Y^T D Y = y_2^2 + 3y_3^2$$

$\therefore$  the quadratic form is positive semi-definite.

$$\text{The transformation } X = NY \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \\ \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} x_1 &= \frac{1}{\sqrt{3}} y_1 + \frac{1}{\sqrt{2}} y_2 - \frac{1}{\sqrt{6}} y_3 \\ x_2 &= \frac{1}{\sqrt{3}} y_1 + \frac{2}{\sqrt{6}} y_3 \\ x_3 &= -\frac{1}{\sqrt{3}} y_1 + \frac{1}{\sqrt{2}} y_2 + \frac{1}{\sqrt{6}} y_3 \end{aligned}$$

These equation make the quadratic form = 0

$$\Rightarrow y_2^2 + 3y_3^2 = 0$$

$$\Rightarrow y_2 = 0 \text{ and } y_3 = 0$$

[Since sum of squares of real numbers = 0  $\Rightarrow$  each number = 0] and  $y_1$  can take any real value.

$$\therefore x_1 = \frac{1}{\sqrt{3}} y_1, \quad x_2 = \frac{1}{\sqrt{3}} y_1, \quad x_3 = -\frac{1}{\sqrt{3}} y_1$$

Choosing  $y_1 = \sqrt{3}$ , one set of values of  $x_1, x_2, x_3$  is

$$x_1 = 1, x_2 = 1, x_3 = -1$$



## EXERCISE 1.3

1. Diagonalise  $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$  by an orthogonal transformation. [AU 2002, 2006]

2. Reduce the matrix  $A = \begin{bmatrix} 10 & -2 & -5 \\ -2 & 2 & 3 \\ -5 & 3 & 5 \end{bmatrix}$  to diagonal form. [AU 2004]

3. Reduce  $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$  to diagonal form by orthogonal reduction. [AU 2007, 2008]

4. Reduce  $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$  to diagonal form by orthogonal reduction.

Hence find  $A^3$ .

5. Diagonalise  $\begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$

6. Find the symmetric matrix  $A$  whose eigen values and eigen vectors are given

(i) eigen values are 0, 2, and eigen vectors  $\begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ .

(ii) eigen values are 1, 2, 3 and eigen vectors  $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

7. Reduce the quadratic form  $8x_1^2 + 7x_2^2 + 3x_3^2 - 12x_1x_2 - 8x_2x_3 + 4x_3x_1$  to the canonical form through an orthogonal transformation and hence show that it is positive semi-definite. [AU 2003]

8. Reduce the quadratic form  $2x_1^2 + 6x_2^2 + 2x_3^2 + 8x_1x_3$  to canonical form by orthogonal reduction. [AU 2005]

9. Reduce the quadratic form  $3x_1^2 + 5x_2^2 + 3x_3^2 - 2x_2x_3 + 2x_3x_1 - 2x_1x_2$  to the canonical form by orthogonal reduction. [AU 2001, 2011]

10. Find the nature, index, signature and rank of the following Q.F, without reducing to canonical form.

(i)  $3x_1^2 + 5x_2^2 + 3x_3^2 - 2x_2x_3 + 2x_3x_1 - 2x_1x_2$ . [AU 2000, 2011]

(ii)  $10x^2 + 2y^2 + 5z^2 + 6yz - 10zx - 4xy$ . [MKU 2000, AU 2010]

(iii)  $3x^2 - 2y^2 - z^2 - 4xy + 8xz + 12yz$ .

**ANSWERS TO EXERCISE 1.3**

1.  $\lambda = 2, 2, 8$ ; eigen vectors  $[0 \ 1 \ 1]^T, [1 \ 1 \ -1]^T, [2 \ -1 \ 1]^T$
2.  $\lambda = 0, 3, 14$ ; eigen vectors  $[1 \ -5 \ 4]^T, [1 \ 1 \ 1]^T, [-3 \ 1 \ 2]^T$
3.  $\lambda = 0, 3, 15$ ; eigen vectors  $[1 \ 2 \ 2]^T, [2 \ 1 \ -2]^T, [2 \ -2 \ 1]^T$
4.  $\lambda = 1, 1, 4$ ; eigen vectors  $[1 \ 1 \ 0]^T, [-1 \ 1 \ 2]^T, [1 \ -1 \ 1]^T$

$$A^3 = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$$

5.  $\lambda = 3, 6, 9$ ; eigen vectors  $[1 \ 2 \ 2]^T, [2 \ 1 \ -2]^T, [2 \ -2 \ 1]^T$
6. (i)  $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$  (ii)  $\begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$
7.  $3y_2^2 + 15y_3^2$
8.  $-2y_1^2 + 6y_2^2 + 6y_3^2$
9.  $2y_1^2 + 3y_2^2 + 6y_3^2$
10. (i) Eigen values are 2, 3, 6; positive definite, index = 3, signature = 3, rank = 3.  
 (ii) Eigen values are 0, 3, 14; positive semi-definite, index = 2, signature = 2, rank = 2.  
 (iii) Eigen values are 3, 6, -9; indefinite, index = 2, signature = 1, rank = 3.

## PART A QUESTIONS AND ANSWERS

1. If  $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ , find the eigen values of  $A^2 - 2A + I$ .

**Ans:** Characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \lambda^2 - 7\lambda + 6 = 0$$

$$\Rightarrow (\lambda - 6)(\lambda - 1) = 0 \Rightarrow \lambda = 6, 1.$$

$\therefore$  the eigen values of  $A^2 - 2A + I$  are  $6^2 - 2 \cdot 6 + 1, 1^2 - 2 \cdot 1 + 1 \Rightarrow 25, 0$

2. Prove that  $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$  and  $-3A^{-1}$  have the same eigen values. [AU 2005]

**Ans:** Characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^2 - 2\lambda - 3 = 0 \Rightarrow (\lambda - 3)(\lambda + 1) = 0 \Rightarrow \lambda = 3, -1$$

$\therefore$  the eigen values of  $-3A^{-1}$  are  $-3\left(\frac{1}{3}\right), -3\left(\frac{1}{-1}\right) \Rightarrow -1, 3$

Hence the eigen values of A and  $-3A^{-1}$  are the same.

3. Two eigen values of the matrix  $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$  are equal to 1 each, find the eigen values of  $A^{-1}$ . [AU 2002]

**Ans:** Given two eigen values of A are 1, 1. Let  $\lambda$  be the third eigen value.

$$\therefore 1 + 1 + \lambda = 2 + 3 + 2 \Rightarrow \lambda = 5$$

$\therefore$  the eigen values of A are 1, 1, 5

$\therefore$  the eigen values of  $A^{-1}$  are 1, 1,  $\frac{1}{5}$

4. If the product of two eigen values of the matrix  $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$  is 2, find the third eigen value. [AU 2008]

**Ans:** Let  $\lambda_1, \lambda_2, \lambda_3$  be the eigen values of A.

Given  $\lambda_1 \lambda_2 = 2$ . We know  $\lambda_1 \lambda_2 \lambda_3 = |A| = 8$

$$\therefore 2\lambda_3 = 8 \Rightarrow \lambda_3 = 4.$$

5. Find the sum and product of the eigen value of the matrix

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \quad [\text{AU 2005}]$$

**Ans:** Sum of the eigen values =  $2 + 2 + 2 = 6$

Product of the eigen values =  $|A| = 2(4 - 0) + 1(-2) = 6$

6. If the sum of two eigen values and trace of a  $3 \times 3$  matrix A are equal, find the value of  $|A|$ . [AU 2005, 2009, 2010]

**Ans:** Let  $\lambda_1, \lambda_2, \lambda_3$  be the eigen values of A

Then  $\lambda_1 + \lambda_2 + \lambda_3 = \text{trace of A}$

But, given  $\lambda_1 + \lambda_2 = \text{trace of A}$

$$\therefore \lambda_1 + \lambda_2 + \lambda_3 = \lambda_1 + \lambda_2 \Rightarrow \lambda_3 = 0$$

$$\therefore |A| = \lambda_1 \lambda_2 \lambda_3 = 0$$

7. Find the constants  $a$  and  $b$  such that the matrix  $\begin{bmatrix} a & 4 \\ 1 & b \end{bmatrix}$  has 3 and  $-2$  as eigen values. [AU 2004]

**Ans:** Let  $A = \begin{bmatrix} a & 4 \\ 1 & b \end{bmatrix}$

Given eigen values are 3,  $-2$

But the eigen values of A are  $a$  and  $b$

$$\therefore a + b = 3 - 2 = 1 \quad (1)$$

$$\text{and } |A| = 3(-2) = -6$$

$$\Rightarrow ab - 4 = -6 \Rightarrow ab = -2 \quad (2)$$

$$(1) \Rightarrow b = 1 - a \quad \therefore a(1 - a) = -2$$

$$\Rightarrow a^2 - a - 2 = 0$$

$$\Rightarrow (a - 2)(a + 1) = 0 \Rightarrow a = 2, -1$$

$$\text{If } a = 2, \text{ then } b = 1 - 2 = -1$$

$$\text{If } a = -1, \text{ then } b = 1 - (-1) = 2$$

$$\therefore a = 2, b = -1 \text{ or } a = -1, b = 2$$

8. Two eigen values of  $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -5 & -2 \end{bmatrix}$  are equal and they are double the third. Find the eigen values of  $A^2$ . [AU 2000]

**Ans:** Let the third eigen value be  $\lambda$ . Then the other two eigen values are  $2\lambda, 2\lambda$

$$\therefore 2\lambda + 2\lambda + \lambda = 4 + 3 + (-2)$$

$$\Rightarrow 5\lambda = 5 \Rightarrow \lambda = 1$$

$$\therefore \text{eigen values of A are } 2, 2, 1$$

$$\therefore \text{eigen of } A^2 \text{ are } 2^2, 2^2, 1^2 \Rightarrow 4, 4, 1.$$

9. If  $\lambda_1, \lambda_2, \dots, \lambda_n$  are the eigen values of an  $n \times n$  matrix  $A$ , then show that  $\lambda_1^3, \lambda_2^3, \dots, \lambda_n^3$  are the eigen values of  $A^3$ . [AU 2003]

**Ans:** Let  $\lambda$  be any eigen value of  $A$ . Then we can find a column matrix  $X \neq 0$  such that

$$AX = \lambda X \quad (1)$$

$$\text{Now } A^2X = A(AX) = A(\lambda X) = \lambda(AX) = \lambda(\lambda X)$$

$$\Rightarrow A^2X = \lambda^2X \quad (2)$$

$$\begin{aligned} \text{and } A^3X &= A(A^2X) = A(\lambda^2X) \\ &= \lambda^2(AX) = \lambda^2(\lambda X) \end{aligned}$$

$$\therefore A^3X = \lambda^3X$$

$\therefore \lambda^3$  is an eigen value of  $A^3$ , by definition.

This is true for all eigen values of  $A$ .

$\therefore \lambda_1^3, \lambda_2^3, \dots, \lambda_n^3$  are eigen values of  $A^3$ .

10. If the eigen values of a  $3 \times 3$  matrix  $A$  are 2, 3, 4, find the eigen values of  $\text{adj } A$ .

**Ans:** The eigen values of  $A$  are 2, 3, 4

$\therefore$  the eigen values of  $A^{-1}$  are  $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$

$$\text{We know } A^{-1} = \frac{\text{adj } A}{|A|} \Rightarrow \text{adj } A = A^{-1}|A|$$

$$\text{But } |A| = \text{product of eigen values} = 2 \cdot 3 \cdot 4 = 24$$

$\therefore$  the eigen values of  $\text{adj } A$  are  $24 \cdot \frac{1}{2}, 24 \cdot \frac{1}{3}, 24 \cdot \frac{1}{4} \Rightarrow 12, 8, 6$

11. The matrix  $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$  has an eigen vector  $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ , find the corresponding eigen value of  $A$ . [AU 2006]

**Ans:** Let  $\lambda$  be the eigen value of  $A$  for which  $X = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$  is an eigen vector.

Then

$$(A - \lambda I)X = 0$$

$$\Rightarrow \begin{bmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -(1-\lambda)+3 \\ -1+1 \\ -3+1-\lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \lambda + 2 = 0 \Rightarrow \lambda = -2$$

$\therefore$  Corresponding eigen value is  $-2$ .

12. Using Cayley-Hamilton theorem find the inverse of  $\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$

[AU 2003]

**Ans:** Let  $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ .

The characteristic equation of A is  $|A - \lambda I| = 0$

$$\Rightarrow \lambda^2 - 4\lambda - 5 = 0 \quad (1)$$

By Cayley-Hamilton theorem A satisfies (1)

$$\therefore A^2 - 4A - 5I = 0$$

$$\Rightarrow 5I = A^2 - 4A$$

$$\text{Multiply by } A^{-1}, \quad 5A^{-1} = A - 4A^{-1}A$$

$$= A - 4I$$

$$= \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{5} \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix}$$

13. If 2, 3 are eigen values of a square matrix A of order 2, express  $A^2$  in terms of A and I. [AU 2007]

**Ans:** Given eigen values of A are 2, 3.

The characteristic equation of A is  $\lambda^2 - S_1\lambda + S_2 = 0$

where  $S_1$  = sum of the diagonal elements

= sum of eigen values

$$= 2 + 3 = 5$$

$$S_2 = |A| = 2 \cdot 3 = 6$$

$\therefore$  the characteristic equation is  $\lambda^2 - 5\lambda + 6 = 0$

A satisfies it, by Cayley-Hamilton theorem

$$\therefore A^2 - 5A + 6I = 0$$

$$\Rightarrow A^2 = 5A - 6I$$

14. If A is an orthogonal matrix, show that  $A^{-1}$  is also orthogonal.

[AU 2004]

**Ans:** Given A is orthogonal.

$$\therefore A^T = A^{-1} = B, \text{ say}$$

$$\text{then } B^T = (A^{-1})^T = (A^T)^{-1}$$

$$\Rightarrow B^T = B^{-1}$$

$\therefore B$  is orthogonal i.e.,  $A^{-1}$  is orthogonal.

**15. Find the nature of the quadratic form  $2x^2 + 2xy + 3y^2$ .**

**Ans:** Matrix of the quadratic form is  $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$

The principal minors are  $D_1 = 2 > 0$ ,  $D_2 = |A| = 5 > 0$

$\therefore$  the Q.F is positive definite.

**16. If the quadratic form  $ax^2 + 2bxy + cy^2$  is positive definite (or negative definite) then prove that the quadratic equation  $ax^2 + 2bx + c = 0$  has imaginary roots.**

**Ans:** The matrix of the quadratic form is  $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$

Then the principle minors are  $D_1 = a$ ,  $D_2 = |A| = ac - b^2$

If the quadratic form is positive definite then  $D_1 > 0$  and  $D_2 > 0$

$$\Rightarrow a > 0 \text{ and } ac - b^2 > 0$$

$$\Rightarrow a > 0 \text{ and } b^2 - ac < 0 \quad (1)$$

If the quadratic form is negative definite then  $(-1)^i D_i > 0 \forall i$

$$\Rightarrow -D_1 > 0, D_2 > 0$$

$$\Rightarrow D_1 < 0, D_2 > 0 \Rightarrow a < 0 \text{ and } ac - b^2 > 0$$

$$\Rightarrow a < 0, b^2 - ac < 0 \quad (2)$$

Now discriminant of the quadratic  $ax^2 + 2bx + c = 0$  is

$$D = 4b^2 - 4ac$$

$$= 4(b^2 - ac) < 0 \quad [\text{by (1) and (2)}]$$

$\therefore$  The roots are imaginary.

**17. Find the nature of the quadratic form  $2x_1x_2 + 2x_1x_3 - 2x_2x_3$ .**

**Ans:** The matrix of the Q.F is  $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}$

The principle minors are  $D_1 = 0$ ,  $D_2 = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1 < 0$ ,

$$D_3 = |A| = -2 < 0$$

$\therefore$  the quadratic form is indefinite.

**18. Find the index and signature of the quadratic form  $x_1^2 + 2x_2^2 - 5x_3^2$ .**

**Ans:** Since only square terms are there, it is in canonical form  $\lambda_1 = 1$ ,  $\lambda_2 = 2$ ,  $\lambda_3 = -5$

$\therefore$  Index = number of + ve eigen values = 2

Signature = difference between the number of positive and negative eigen values =  $2 - 1 = 1$

**19. What is the sum of the squares of the eigen values of** 
$$\begin{bmatrix} 1 & 7 & 5 \\ 0 & 2 & 9 \\ 0 & 0 & 5 \end{bmatrix}$$

[AU 2001, 2010]

**Ans:** We know the eigen values of a triangular matrix are the diagonal elements of the matrix.

$\therefore$  the eigen values are 1, 2, 5

$\therefore$  sum of the squares =  $1^2 + 2^2 + 5^2 = 30$

**20. Write down the matrix of the quadratic form**

$$2x^2 + 8z^2 + 4xy + 10xz - 2yz.$$

[AU 2001]

**Ans:** The given quadratic form is

$$2x^2 + 0y^2 + 8z^2 + 2(2)xy + 2(5)xz + 2(-1)yz$$

$$\therefore \text{ the matrix is } A = \begin{bmatrix} 2 & 2 & 5 \\ 2 & 0 & -1 \\ 5 & -1 & 8 \end{bmatrix}$$

**21. For a given matrix A of order 3,  $|A| = 32$  and two of its eigen values are 8 and 2. Find the sum of the eigen values**

[AU 2009]

**Ans:** Given  $|A| = 32$  and two of the eigen values are 8 and 2.

Since A is a  $3 \times 3$  matrix, it has another eigen value.

Let it be  $\lambda$ . We know product of the eigen values =  $|A|$

$$\text{But } |A| = 32$$

$$\Rightarrow 8 \cdot 2\lambda = 32 \Rightarrow \lambda = 2$$

**22. Check whether matrix B is orthogonal? Justify.**

$$B = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

[AU 2009]

**Ans:** Given  $B = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$



$$\therefore B^T = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned} \text{Now } BB^T &= \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & -\cos \theta \sin \theta + \sin \theta \cos \theta & 0 \\ -\sin \theta \cos \theta + \sin \theta \cos \theta & \sin^2 \theta + \cos^2 \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \end{aligned}$$

$$\begin{aligned} \text{and } B^T B &= \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & \cos \theta \sin \theta - \sin \theta \cos \theta & 0 \\ \sin \theta \cos \theta - \sin \theta \cos \theta & \sin^2 \theta + \cos^2 \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \end{aligned}$$

$$BB^T = B^T B = I \quad \therefore B \text{ is orthogonal.}$$

**23. Use Cayley-Hamilton theorem to find  $A^4 - 4A^3 - 5A^2 + A + 2I$  when**

$$A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \quad \text{[AU 2009]}$$

**Ans:** Given  $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$

$$\therefore \text{characteristic equation is } |A - \lambda I| = 0$$

$$\Rightarrow \begin{bmatrix} 1-\lambda & 2 \\ 4 & 3-\lambda \end{bmatrix} = 0$$

$$\Rightarrow (1-\lambda)(3-\lambda) - 8 = 0$$

$$\Rightarrow \lambda^2 - 4\lambda + 3 - 8 = 0$$

$$\Rightarrow \lambda^2 - 4\lambda - 5 = 0 \quad (1)$$

$\therefore$  By Cayley-Hamilton theorem

A satisfies (1)

$$\therefore A^2 - 4A - 5I = 0 \quad (2)$$

To find  $A^4 - 4A^3 - 5A^2 + A + 2I$

$$A^4 - 4A^3 - 5A^2 + A + 2I = A^2(A^2 - 4A - 5I) + (A + 2I)$$

$$\Rightarrow A^4 - 4A^3 - 5A^2 + A + 2I = A + 2I \quad [\text{Using (2)}]$$

$$= \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} + 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 5 \end{bmatrix}$$

- 24. True or false: If A and B are two invertible matrices then AB and BA have the same eigen values.** [AU 2009, 2011]

**Ans:** True. Given A and B are invertible.

$\therefore A^{-1}$  and  $B^{-1}$  exist.

$$\text{Now } A^{-1}(AB)A = (AA^{-1})BA = IBA = BA.$$

So, AB and BA are similar.

We know that similar matrices have the same eigen values.

$\therefore$  AB and BA have same eigen values.

- 25. If the sum of the eigen values of the matrix of the quadratic form is equal to zero, then what will be the nature of the quadratic form?** [AU 2009]

**Ans:** If the quadratic form is in two variables, then the matrix of the quadratic form is a  $2 \times 2$  matrix.

Let  $\lambda_1, \lambda_2$  be the eigen values.

If  $\lambda_1 + \lambda_2 = 0$ , then  $\lambda_1$  and  $\lambda_2$  have opposite signs.

So, the quadratic form is indefinite.

- 26. A is a singular matrix of order three, 2 and 3 are the eigen values. Find the third eigen value.** [AU 2010]

**Ans:** A is  $3 \times 3$  matrix.

Given  $|A| = 0$  and 2 and 3 are two eigen values.

Let  $\lambda_3$  be the third eigen value.

But the product of the eigen values  $= |A| = 0$

$$\Rightarrow 2 \times 3 \times \lambda_3 = 0$$

$$\Rightarrow \lambda_3 = 0$$

- 27. Find the eigen vector corresponding to the eigen value 1 of the matrix**

$$A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

[AU 2009]

**Ans:** Given 1 is the eigen value of the matrix  $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$

Let  $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$  be the eigen vector corresponding to the eigen value 1.

$$\begin{aligned}
 \therefore (A - \lambda I) &= 0 \Rightarrow (A - I)X = 0 \\
 \Rightarrow \begin{bmatrix} 2-1 & 2 & 1 \\ 1 & 3-1 & 1 \\ 1 & 2 & 2-1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
 \Rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
 \Rightarrow x_1 + 2x_2 + x_3 &= 0
 \end{aligned}$$

Choose  $x_3 = 0$ , then  $x_1 + 2x_2 = 0 \Rightarrow x_1 = -2x_2$

Choose  $x_2 = 1$ ,  $\therefore x_1 = -2$

$\therefore$  an eigen vector corresponding to the eigen value 1 is  $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$

**28. Find the sum and product of the eigen values of the matrix**

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 0 & 3 \\ -2 & -1 & -3 \end{bmatrix}. \quad [\text{AU 2010}]$$

**Ans:** Sum of the eigen values = Sum of the diagonal elements of A  
 $= 1 + 0 - 3 = -2$

Product of the eigen values =  $|A|$

$$\begin{aligned}
 &= \begin{vmatrix} 1 & 2 & -2 \\ 1 & 0 & 3 \\ -2 & -1 & -3 \end{vmatrix} \\
 &= -1(-6 - 2) - 3(-1 + 4) \\
 &= 8 - 9 = -1 \quad [\text{expanding by the second row}]
 \end{aligned}$$

**29. Find the nature of the quadratic form  $2x_1^2 - x_2^2$ .** [AU 2009, 2010]

**Ans:** Q.F is  $2x_1^2 - x_2^2$

Since only square terms are there, it is in Canonical form.

There are positive and negative terms and hence the Q.F is indefinite.

**30. If 2 is an eigen value of  $A = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$  find the other eigen values.** [AU 2010]

**Ans:** Given 2 is an eigen value of A. Let  $\lambda_2, \lambda_3$  be the other eigen values of A.

$\therefore 2 + \lambda_2 + \lambda_3 = \text{Sum of the diagonal elements of A.}$

$$\begin{aligned}\Rightarrow 2 + \lambda_2 + \lambda_3 &= 2 + 1 - 1 \\ \Rightarrow \lambda_2 + \lambda_3 &= 0 \\ \Rightarrow \lambda_2 &= -\lambda_3\end{aligned}\tag{1}$$

We have  $2\lambda_2\lambda_3 = |A|$

$$\begin{aligned}&= \begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 4 & -2 \end{bmatrix} \begin{array}{l} C_2 \rightarrow C_2 + C_1 \\ C_3 \rightarrow C_3 - C_1 \end{array} \\ &= 2 \cdot 2 \cdot (-2) = -8\end{aligned}$$

$$\Rightarrow 2 \cdot (-\lambda_3)\lambda_3 = -8 \Rightarrow \lambda_3^2 = 4 \Rightarrow \lambda_3 = \pm 2$$

If  $\lambda_3 = 2$ ,  $\lambda_2 = -2$  and if  $\lambda_3 = -2$ ,  $\lambda_2 = 2$

$\therefore$  the other two eigen values are  $-2, 2$

**31. Discuss the nature of the following quadratic form**

$$3x_1^2 + 3x_2^2 - 5x_3^2 - 2x_1x_2 - 6x_1x_3 - 6x_3x_1.$$

[AU 2010]

**Ans:** The matrix of the Q.F is  $A = \begin{bmatrix} 3 & -1 & -3 \\ -1 & 3 & -3 \\ -3 & -3 & -5 \end{bmatrix}$

The principal minors are  $D_1 = 3 > 0$ ,

$$D_2 = \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} = 9 - 1 = 8 > 0$$

and

$$\begin{aligned}D_3 &= \begin{vmatrix} 3 & -1 & -3 \\ -1 & 3 & -3 \\ -3 & -3 & -5 \end{vmatrix} \\ &= 3(-15 + 9) + 1(5 - 3) + (-3)(-3 + 3) \\ &= -18 + 2 + 0 = -16 < 0\end{aligned}$$

Since  $D_1 > 0$ ,  $D_2 > 0$  and  $D_3 < 0$ , the Q.F is indefinite.

**32. Two of the eigen values of  $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$  are 3 and 6. Find the eigen values of  $A^{-1}$ .**

[AU 2010]

**Ans:** Let  $\lambda_1, \lambda_2, \lambda_3$  be the three eigen values of A.

$$\text{Let } \lambda_1 = 3, \lambda_2 = 6$$

$\therefore \lambda_1 + \lambda_2 + \lambda_3 = \text{Sum of the diagonal elements of A.}$

$$\Rightarrow 3 + 6 + \lambda_3 = 3 + 5 + 3 \Rightarrow \lambda_3 = 2$$

$\therefore$  the eigen values of A are 2, 3, 6.

Hence the eigen values of  $A^{-1}$  are  $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$

**33. Find the sum of the squares of the eigen values of the matrix**

$$A = \begin{bmatrix} 1 & 7 & 5 \\ 0 & 2 & 9 \\ 0 & 0 & 5 \end{bmatrix}. \quad [\text{AU 2010}]$$

**Ans:** Since A is upper triangular matrix, the eigen values are the diagonal elements of A. i.e., the eigen values of A are 1, 2, 5.

$\therefore$  sum of the squares of the eigen values  $= 1^2 + 2^2 + 5^2 = 30$

**34. State the nature of quadratic form  $2xy + 2yz + 2zx$ .** [AU 2010]

**Ans:** The matrix of the quadratic form is  $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$$\therefore D_1 = 0, D_2 = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1 < 0$$

$$\text{and } D_3 = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = 0 - 1(0 - 1) + 1 \cdot (1 - 0) = 2$$

Since  $D_1 = 0, D_2 < 0, D_3 > 0$ , the nature of the quadratic form is indefinite.

**35. Write down the matrix of the quadratic form  $Q(x, y) = 3x^2 + 2y^2 - 4xy$ .** [AU 2010]

**Ans:** The quadratic form is  $3x^2 + 2y^2 - 4xy$

$$\therefore \text{ the matrix of the quadratic form is } A = \begin{bmatrix} 3 & -2 \\ -2 & 2 \end{bmatrix}$$

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